

# **MINIMIZATIONS OF LEAKAGE CURRENT BY DETERMINING FREQUENCY RANGE IN TRANSFORMER-LESS PV INVERTERS**

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**DEPARTMENT OF ELECTRICAL AND ELECTRONIC  
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GOPALGANJ - 8100**

*September, 2025*

## **DEDICATION**

This research is fully dedicated to our parents, who have consistently provided us with financial, spiritual, and moral support, and who have inspired us and given us strength when we felt like giving up. We extend our gratitude to our siblings, family members, mentors, friends, and classmates who provided guidance and support throughout the completion of this study. Finally, this book is dedicated to the Almighty God, whom we thank for providing us with a healthy life, wisdom, mental strength, protection, and abilities. We provide you all of these.

# CERTIFICATE

This is to certify that the thesis entitled “**Minimizations of Leakage Current by Determining Frequency Range in Transformer-less PV Inverters**” being submitted by the following students:

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# DECLARATION

We hereby declare that the thesis entitled “**Minimizations of Leakage Current by Determining Frequency Range in Transformer-less PV Inverters**” has been carried out by us under the supervision of **Dr. A.T.M. Saiful Islam**, Associate Professor, Department of Electrical and Electronic Engineering, Gopalganj Science and Technology University. This work has not been submitted elsewhere for any other degree or diploma.

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# ABSTRACT

Due to the economic advantage and the relatively lesser negative impact on the environment, the use of distributed energy resources and photovoltaic systems (PV systems) is rapidly increasing. Photovoltaic system inverters, which are fitted with transformers, do offer galvanic isolation between grid and pv system. However, they are less efficient, size-inefficient, and high-cost. There is particular use of transformer-less inverters, as they are more efficient, cheaper, and smaller in size compared to the transformer-based inverter. However, they are at risk of more leakage current as they do not provide galvanic isolation. This paper investigates the use of Full Bridge Four Switch (H4) and Highly Efficient and Reliable Inverter Concept (HERIC) inverters and how they deal with the leakage current, common mode voltage, differential mode voltage, and Total Harmonic Distortion (THD) developed at various frequency ranges. Simulations were run for each design using MATLAB (Simulink) versions R2025b and R2018a. The HERIC topology shows the best performance in terms of leakage current, low THD, and high efficiency. The data shows that the HERIC Unipolar topology has a leakage current of 40.27 mA at 26 kHz, having an efficiency of 98.97%. On the other hand, the HERIC Bipolar architecture has a leakage current that is about 15 times lower than that of the Unipolar HERIC. In relation to the H4 unipolar topology, which has a leakage current about 4 times higher than HERIC bipolar, surpassing the HERIC unipolar performance, but is inferior to the HERIC bipolar. The HERIC Bipolar topology is especially beneficial for high-voltage, grid-connected PV systems, delivering low leakage current, high efficiency, and minimal harmonic distortion.

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# Abbreviations

|               |                                                |
|---------------|------------------------------------------------|
| PV            | Photovoltaic                                   |
| MPPT          | Maximum Power Point Tracking                   |
| DC            | Direct Current                                 |
| AC            | Alternating Current                            |
| THD           | Total Harmonic Distortion                      |
| EMI           | Electromagnetic Interference                   |
| RMS           | Root Mean Square                               |
| FFT           | Fast Fourier Transform                         |
| CMV           | Common Mode Voltage                            |
| CMI           | Common Mode Interference                       |
| PWM           | Pulse Width Modulation                         |
| SPWM          | Sinusoidal Pulse Width Modulation              |
| SVPWM         | Space Vector Pulse Width Modulation            |
| LCL           | Inductor Capacitor Inductor filter             |
| LC            | Inductor Capacitor filter                      |
| L             | Inductor filter                                |
| PI            | Proportional-Integral                          |
| PID           | Proportional-Integral-Derivative               |
| VSI           | Voltage Source Inverter                        |
| HERIC         | Highly Efficient and Reliable Inverter Concept |
| H5, H6, HERIC | Transformer-less inverter topologies           |
| RCD           | Residual Current Device                        |
| IEC           | International Electro-technical Commission     |
| NPC           | Neutral Point Clamped                          |
| EMI           | Electromagnetic Interference                   |



# Notations

|            |                                                   |
|------------|---------------------------------------------------|
| $V_{dc}$   | DC link voltage                                   |
| $V_{ac}$   | AC grid voltage                                   |
| $I_g$      | Grid current                                      |
| $I_l$      | Leakage current                                   |
| $C_m$      | Parasitic capacitance between PV array and ground |
| $f_s$      | Switching frequency                               |
| $f_g$      | Grid frequency                                    |
| $L, C$     | Inductor and Capacitor values in filter           |
| $R$        | Resistance                                        |
| $\eta$     | Efficiency                                        |
| $\phi$     | Phase angle                                       |
| $\omega_g$ | Angular frequency of grid ( $2\pi f_g$ )          |
| $\omega_c$ | Angular cut-off frequency ( $2\pi f_c$ )          |
| $Z(s)$     | Impedance in Laplace domain                       |

# Chapter 1

## Introduction

### 1.1 Background of Renewable Energy and Photovoltaic Systems

Growing worldwide energy demand and environmental concerns have accelerated the transition from fossil fuels to renewable energy sources in recent decades. Solar photovoltaic (PV) technology is one of the most dependable and rapidly expanding renewable energy sources (figure 1.1). PV systems use semiconductor materials to directly convert sunlight into electrical energy, providing a scalable and sustainable power generation solution. Large-scale grid integration of solar energy is now feasible due to supportive government policies and the falling costs of PV modules.

### 1.2 On-Grid PV Systems and the Role of Inverters

Inverters are essential components of on-grid photovoltaic systems because they transform the direct current (DC) produced by solar panels into alternating current (AC) that may be supplied to the utility grid (figure 1.2). Besides simple conversion, inverters are also responsible for maintaining power quality, grid synchronization, and ensuring compliance with grid codes. The total performance of the photovoltaic system is determined mainly by the efficiency of the inverter. Traditionally, transformer-based inverters were widely used due to their ability to provide galvanic isolation and ensure safety. However, they are bulky, expensive, and less efficient compared to transformerless inverters.[25]

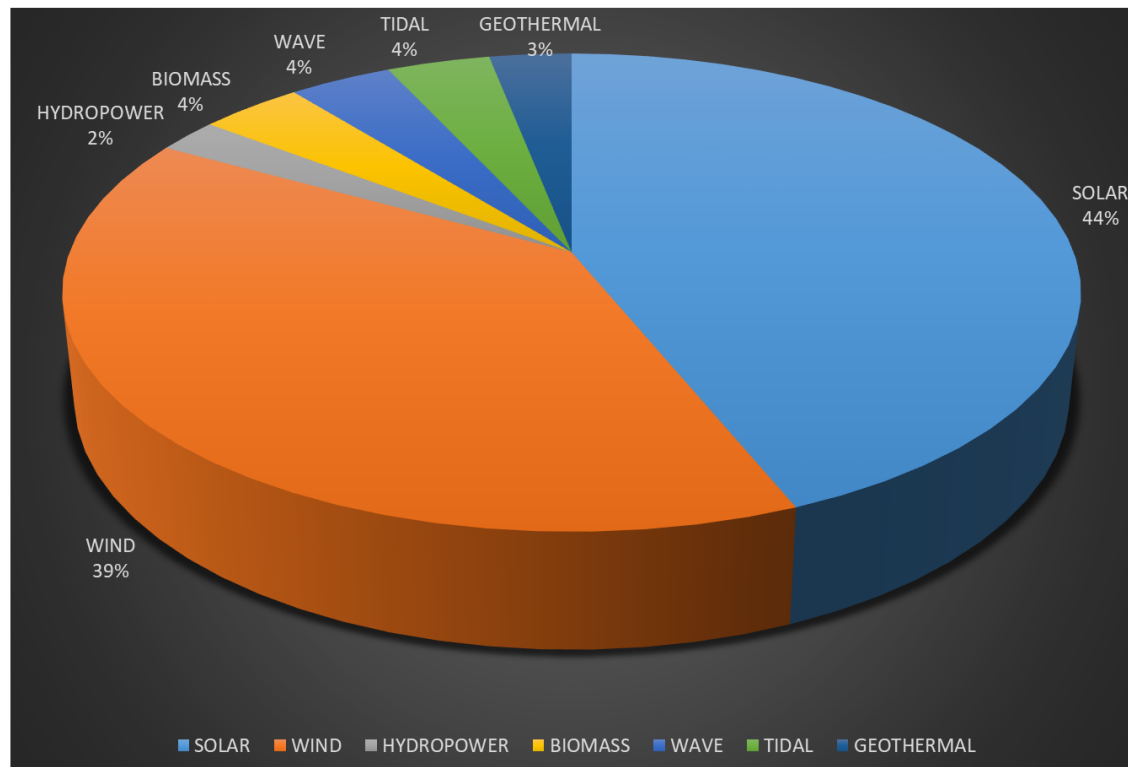


Figure 1.1: Global renewable energy contribution with a focus on PV growth.

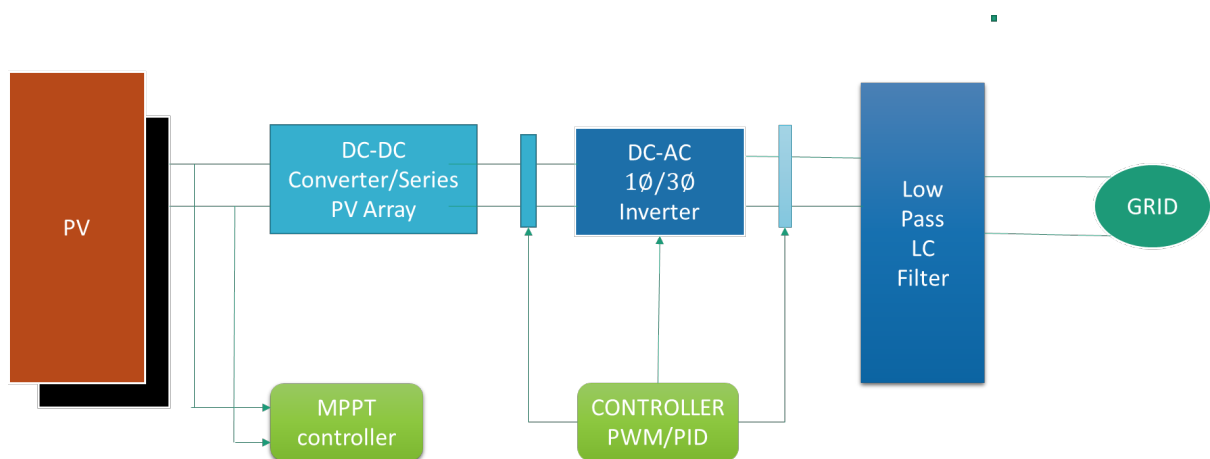


Figure 1.2: Block diagram of a grid-connected PV system highlighting the inverter stage.

### 1.3 Transformerless Inverters: Advantages and Challenges

Transformerless inverters, as shown in 1.3, have garnered significant attention due to their affordability, compact dimensions, and high efficiency. By removing the line-frequency transformer, both power losses and system weight are reduced, resulting in an overall efficiency improvement of 1–2%. Nonetheless, the lack of galvanic isolation presents difficulties, especially concerning common-mode voltage (CMV) and leakage current. These effects arise due to parasitic capacitance between the PV panel and ground, which can compromise system safety and reliability if not properly controlled. (see Appendix A.1) [31]

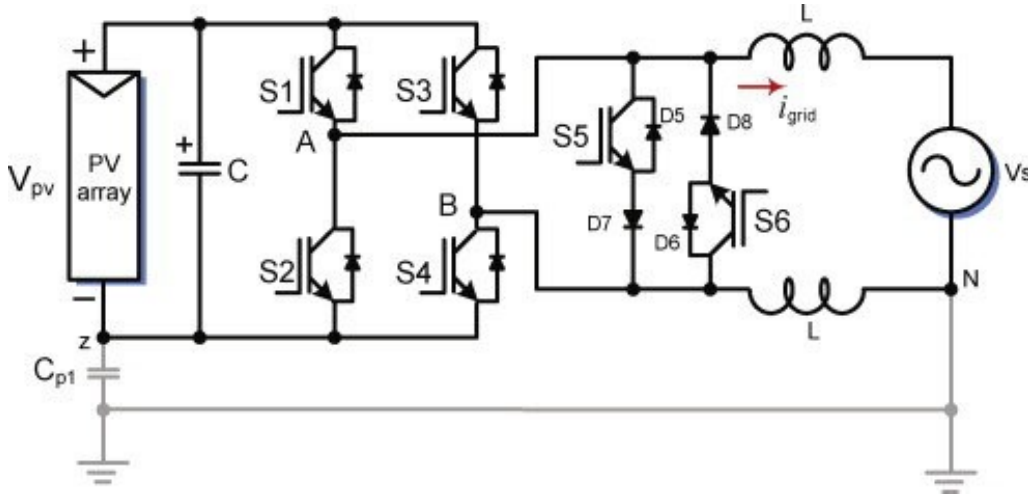


Figure 1.3: Transformerless inverter architectures (HERIC topology) [31].

### 1.4 Leakage Current in PV Systems

The parasitic capacitance between the PV array and the ground is what causes most of the leakage current in PV systems that don't use transformers. This current can take unintended paths that can cause problems like electromagnetic interference (EMI), lower efficiency, possible dangers to maintenance workers, and violations of grid safety rules. The inverter's topology, switching frequency, modulation system, and grounding configuration all have an effect on how much leakage current there is. In order to design a transformerless inverter, it's very important to lower the leakage current. [35] (see Appendix A.2)

**Parasitic Capacitance:** In Figure 1.5, the parasitic capacitance occurs between the photovoltaic panel and the ground as a result of the module's physical design and its

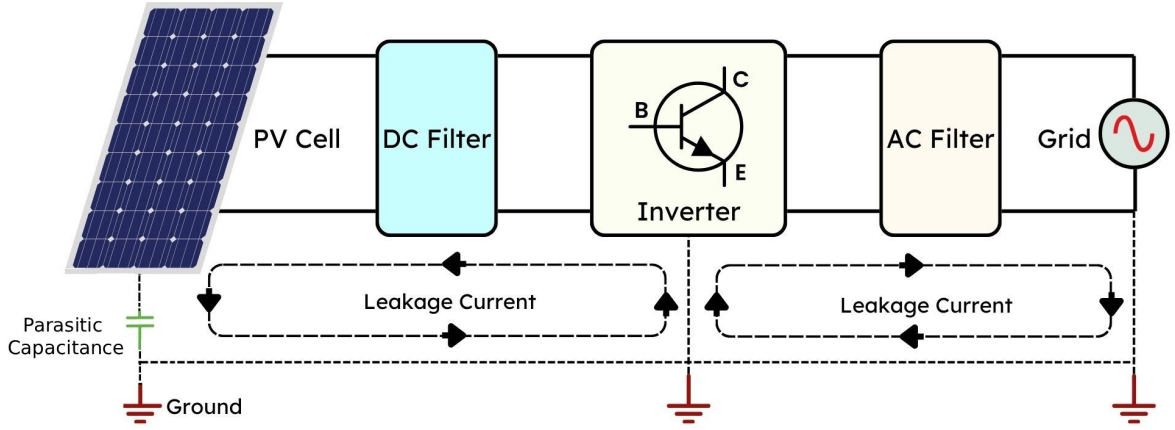


Figure 1.4: Leakage Current in PV Systems

insulating layers. It is essential in influencing the behavior of leakage current in transformerless inverters (see Appendix A.3 for the formal definition).

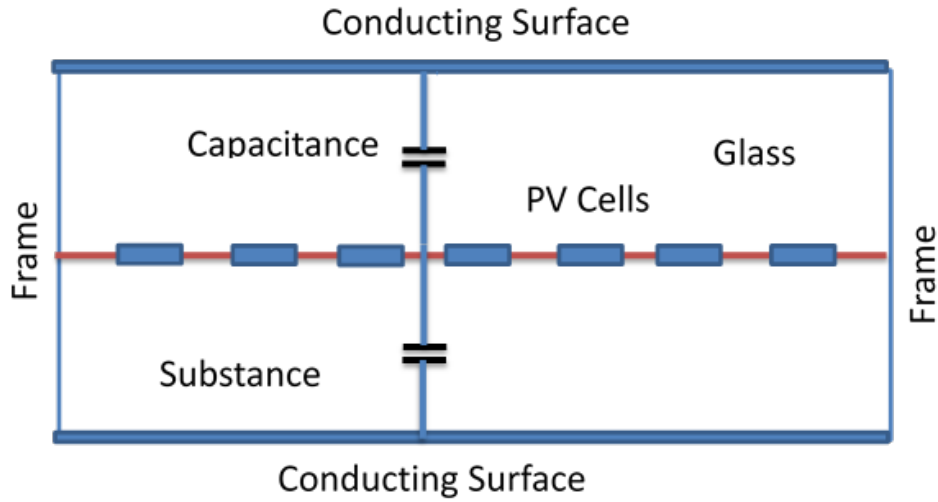


Figure 1.5: Parasitic capacitance in the PV cell.

## 1.5 Significance of Switching Frequency

Efficiency, which is influenced by the switching frequency, is one of the main functions of a transformer-less inverter. Although passive parts like filters can be made smaller by using higher switching frequencies, power devices may experience more switching losses and thermal stress. Additionally, the spectral distribution of harmonics and the behavior of leakage current are directly impacted by switching frequency. Thus, finding the ideal frequency range is crucial to striking a balance between power quality, efficiency, and leakage current reduction.

## 1.6 Research Background and Problem Statement

The H4, H5, H6, HERIC, and Neutral-Point Clamped (NPC) inverters are some of the different types of inverters that can help with leakage current problems in grid-connected photovoltaic (PV) systems. Even if these topologies offer solutions, it is still very challenging to lower leakage current, efficiency, and total harmonic distortion (THD) all at the same time in the field of PV systems. Researchers have looked into using unipolar and bipolar Pulse Width Modulation (PWM) management schemes to cut down on leakage currents. However, it is challenging to accomplish optimal performance while simultaneously reducing these variables, as indicated in prior research [1]; [2]. The results reveal that the HERIC Bipolar topology has the least leakage current and works best for photovoltaic systems. It is especially useful for high-voltage, grid-connected PV systems since it has low leakage, high efficiency, and little harmonic distortion.

A major flaw in earlier studies is that they did not investigate the optimal switching frequency that reduces leakage current while simultaneously enhancing overall efficiency. Most studies, like [4] and [7], focus on specific topologies or control methods, but they don't look at the best switching frequency for transformerless systems that balances performance and safety.

The originality of this paper resides in its comprehensive analysis of the correlation between switching frequency and leakage current across various inverter topologies. The goal of this study is to identify the frequency range that works best for minimizing leakage current while maximizing overall efficiency and making sure that solar systems can be safely and reliably integrated into the grid. This paper presents a comparative analysis of leakage current suppression across various topologies, including HERIC and H4, at different switching frequencies, thereby offering novel insights into their appropriateness for high-performance photovoltaic applications, in contrast to prior studies.

As noted in previous studies [8], many methods and topologies have been suggested, but there is still no agreement on the best way to stop leakage current while also making sure that the system works well in a wide range of conditions.

Leakage current is a major obstacle to the safe and effective operation of transformerless inverters, which are commonly used in grid-connected photovoltaic systems (Figure 1.6). However, the switching frequency is typically chosen based on the design's practical performance rather than a thorough analysis of its leakage performance. To guarantee the effective, secure, and dependable integration of solar systems into the grid, this study attempts to minimize leakage current while identifying the ideal frequency range.

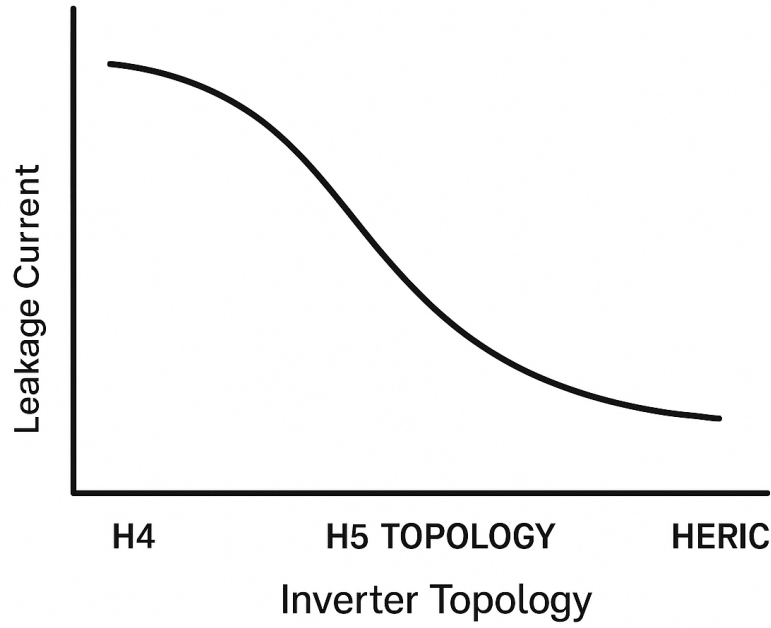


Figure 1.6: Common transformerless inverter topologies with leakage current control strategies.

## 1.7 Challenges and Limitations

The following challenges and limitations were encountered during the course of this research:

- In Bangladesh, exercising research on this topic is not well-established, making it difficult to find relevant research studies on this subject.
- Simulation of this topic using MATLAB techniques is not readily available online or in other sources to learn, study, and analyze.
- Leakage current analysis in simulation is difficult to perform and maintain when considering the switching frequency.
- A high-configuration laptop is necessary for running paid software like Overleaf and MATLAB simulation, which adds an additional challenge.
- We commit to performing thermal evaluation, but the existing learning capabilities of the Simscape tutorial are not suitable, posing a significant limitation to this chapter.
- In LaTeX format, the compiler limit is exceeded in the free mode; paid applications are needed for further extension.
- Power calculation in HERIC bipolar systems is difficult due to the presence of high-amplitude DC current.
- Inconsistent guidelines due to the lack of available research in this field within the

country.

- The implementation of practical projects is limited due to the inconsistency of materials and components available in the lab.
- The output waveform was filtered using a basic low-pass filter, however, additional development of the filtering environment is needed to obtain more effective leakage current isolation. Furthermore, using advanced filtering methods on various topologies can be an interesting alternative to improve the leakage current reduction and the overall system performance, which will be explored in our future work.

## **1.8 Objectives of the Thesis**

The main objectives of this study are as follows:

- To investigate how leakage current behaves in photovoltaic systems with transformerless inverter topologies.
- To investigate the effect of switching frequency on leakage current and inverter efficiency.
- To determine an optimal frequency range that minimizes leakage current without compromising power quality.
- To validate the proposed approach using simulation and experimental analysis.
- To investigate the effect of switching frequency on inverter efficiency.



# Chapter 2

## Types of $1\phi$ Transformer-less Inverter Topologies

### 2.1 Introduction

A power electronics system that is used to change DC input to AC output is called an inverter. Hopefully the DC input and desired AC output magnitude would be the same, but in reality inverters lose some power via switching losses, internal relaxation, or environmental issues, and so you would get a smaller AC signal out compared to what went in. Inverters do not create power, the output power is adjusted according to the DC supply.

Inverters are Voltage Source Inverters (VSI), which need constant voltage, or Current Source Inverters (CSI), which need constant current. CSI produces single or three-phase adjustable AC, whereas VSI generates single or three-phase of adjustable amounts and frequency. Inverters may be utilized in single phase or three-phase electrical systems, as required by the system.

While single-phase inverters generally serve residential needs and can feed into the utility grid if household requirements are met, three-phase inverters are primarily used in commercial applications to supply three-phase loads. Various modulation techniques are employed for switching inverter switches, influencing efficiency, Total Harmonic Distortion (THD), leakage current, and Electromagnetic Interference (EMI). For instance, unipolar, bipolar, and hybrid modulation are used in single-phase full-bridge inverters, with unipolar modulation often preferred for its higher efficiency and lower core loss. Transformer types, low-frequency (LF) and high-frequency (HF), are chosen based on

PV module configuration. LF transformers are heavy, bulky, highly costly, also comparatively less efficient due to core and copper power loss. While single-phase inverters generally serve residential needs and can feed into the utility grid if household requirements are met, three-phase inverters are primarily used in commercial applications to supply three-phase loads.

To overcome these limitations and achieve higher system efficiency, transformer-less inverter topologies have been introduced, which are also smaller, lighter, and less costly. Power electronics and various electrical components are needed to implement a single-phase transformer-less inverter topologies.

## 2.2 Classification of Single-Phase Transformer-less Inverter Topologies

Because of their smaller size, greater efficiency, and lower cost, voltage source inverters (VSIs) are recommended for photovoltaic (PV) systems over current source inverters (CSIs). To comply with grid standards. In order to meet grid standards and enhance the performance of grid-connected PV systems, numerous single-phase transformer-less VSI topologies have been created. Within the single-input category, these topologies are categorized by how they handle common-mode current ( $i_{cm}$ ), such as buck-boost, common ground, H-bridge, and H6 arrangements, as well as by the needed DC-link voltage (either  $2 * V_{PV}$  or  $V_{PV}$ ).

From the above types of topologies, we will explain the H4, H5, H6, and HERIC topologies.

### 2.2.1 H4 Inverter Topology

The H4 inverter is a transformer-less single-phase full-bridge topology widely used for grid-connected photovoltaic (PV) systems. It consists of four active switches (S1–S4) with their antiparallel diodes (D1–D4), as illustrated in Figure 2.2. The topology can be operated under three modulation strategies: unipolar, bipolar, or hybrid. Each scheme impacts efficiency, leakage current, and common-mode (CM) voltage differently (see Appendix A.13.1 for the formal definition).

**Modulation Techniques:**

## Types of $1\phi$ Transformer-less Inverter Topologies

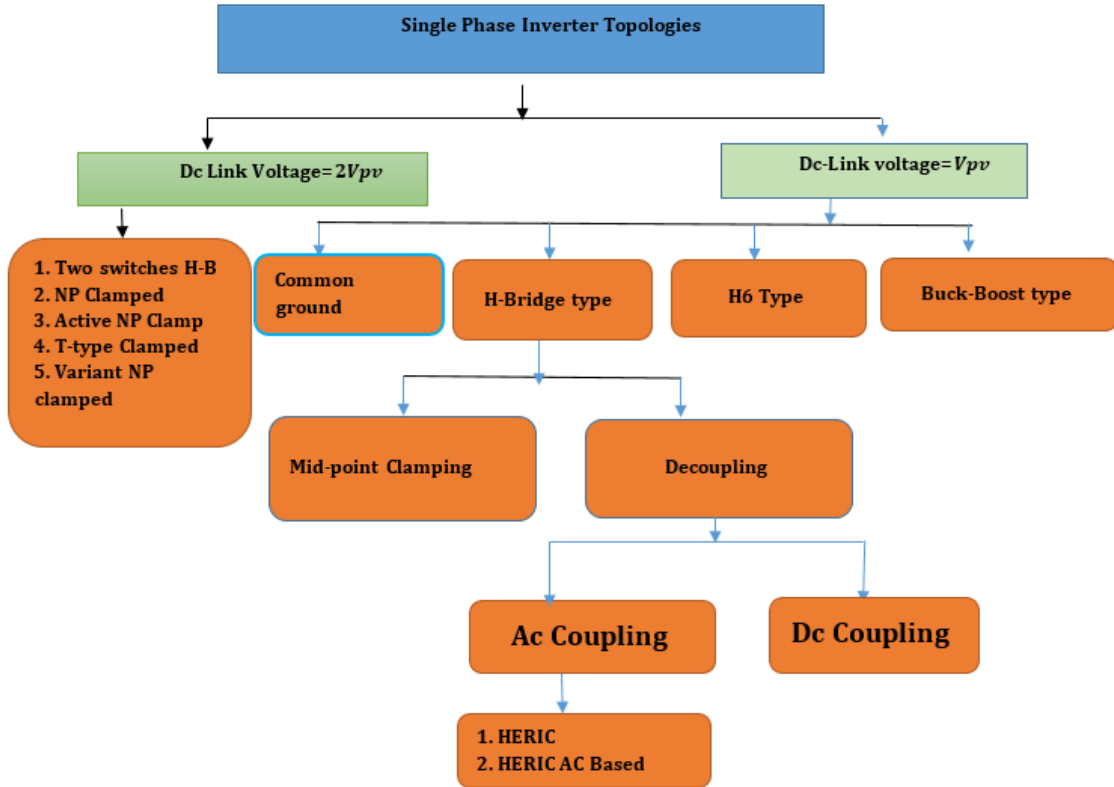


Figure 2.1: Classification of single-phase transformer-less inverter topologies used in PV systems.

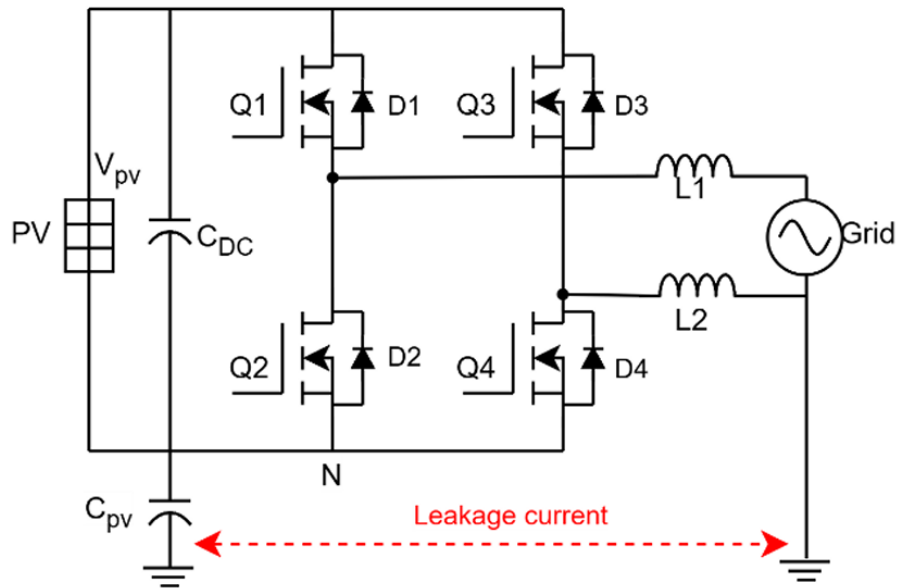


Figure 2.2: H4 inverter circuit topology [2].

- **Unipolar Modulation:** Produces three distinct output voltage levels ( $+V_{pv}$ ,  $0$ ,  $-V_{pv}$ ). It minimizes core losses and improves efficiency but introduces high-frequency common-mode voltage, which increases leakage current.
- **Bipolar Modulation:** Produces only two output voltage levels ( $+V_{pv}$  and  $-V_{pv}$ ). This results in higher filter and switching losses but maintains constant CM voltage, thereby reducing leakage current.
- **Hybrid Modulation:** A trade-off that combines the benefits of both methods. It reduces core losses and leakage current simultaneously by introducing intermediate switching states.

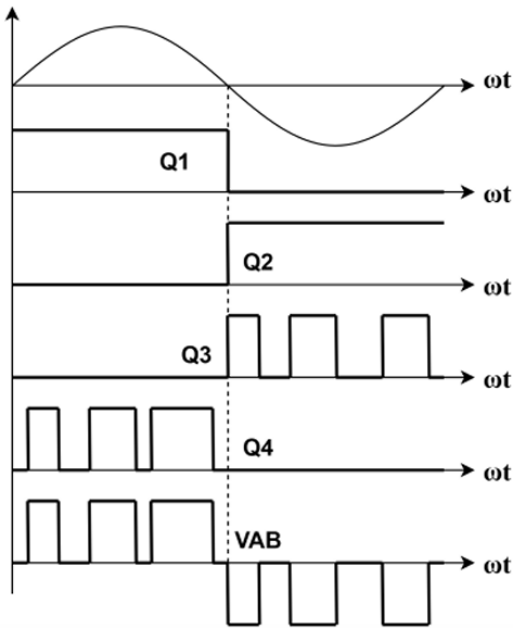


Figure 2.3: Unipolar switching pattern in H4 topology.

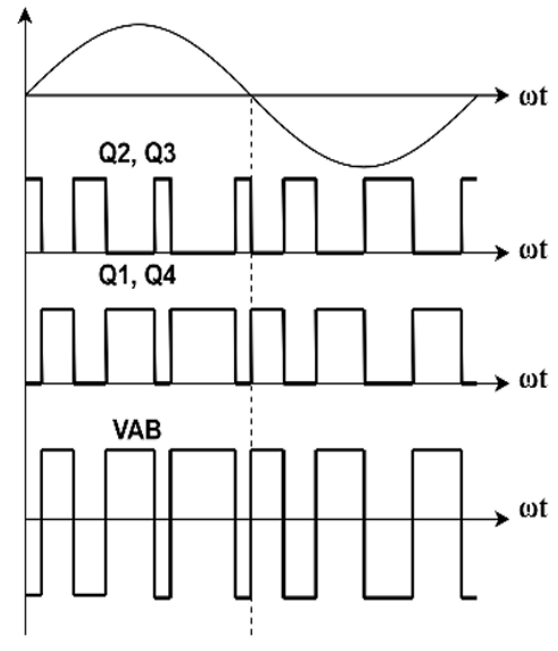


Figure 2.4: Bipolar switching pattern in H4 topology.

[2]

### Modes of Operation in Unipolar Modulation

In unipolar modulation, the H4 inverter operates in four distinct modes depending on the switching states. These modes determine whether energy is delivered from the PV source to the grid or whether the inductor releases its stored energy during freewheeling. Modes 1 and 3 correspond to energy delivery, while Modes 2 and 4 are freewheeling intervals. The inverter output is characterized by the differential mode voltage (DMV), while the common-mode voltage (CMV) depends on the average of the pole voltages  $V_{AN}$  and  $V_{BN}$ .

**Mode 1: Energy Delivery (Positive Half-Cycle)** Switches  $Q_1$  and  $Q_4$  are ON, while  $Q_2$  and  $Q_3$  are OFF. Current flows directly from the PV source to the grid through the

## *Types of 1 $\phi$ Transformer-less Inverter Topologies*

inverter legs.

$$\begin{aligned}\text{CMV} &= \frac{V_{AN} + V_{BN}}{2} = \frac{V_{pv} + 0}{2} = \frac{V_{pv}}{2} \\ \text{DMV} &= V_{AN} - V_{BN} = V_{pv}\end{aligned}$$

Here, the inverter output voltage equals the full PV voltage, and power is actively delivered to the grid.

**Mode 2: Freewheeling (Positive Half-Cycle)** Switch  $Q_1$  remains ON, while current freewheels through the diode  $D_3$  connected to  $Q_3$ . This allows the inductor to release stored energy to the grid.

$$\text{CMV} = \frac{V_{AN} + V_{BN}}{2} = V_{pv}, \quad \text{DMV} = V_{AN} - V_{BN} = 0$$

The inverter output is zero, but the grid current remains continuous due to inductor energy release.

**Mode 3: Energy Delivery (Negative Half-Cycle)** Switches  $Q_2$  and  $Q_3$  are ON, while  $Q_1$  and  $Q_4$  are OFF. Current now flows in the reverse direction from PV to the grid.

$$\text{CMV} = \frac{0 + V_{pv}}{2} = \frac{V_{pv}}{2}, \quad \text{DMV} = V_{AN} - V_{BN} = -V_{pv}$$

The inverter output is negative, corresponding to the negative half-cycle of the AC waveform.

**Mode 4: Freewheeling (Negative Half-Cycle)** Switch  $Q_2$  is ON, and the current freewheels through diode  $D_4$  of  $Q_4$ . Similar to Mode 2, the inductor releases its stored energy.

$$\text{CMV} = 0, \quad \text{DMV} = 0$$

During this interval, the output voltage is zero, but the inductor ensures smooth current delivery.

In summary, unipolar modulation enables four distinct operating states: two active energy transfer modes and two freewheeling modes. This allows the H4 inverter to generate three voltage levels ( $+V_{pv}$ ,  $0$ ,  $-V_{pv}$ ), reducing filter size and improving efficiency. However, the alternating CMV introduces leakage currents, which is an important design consideration.[7]

## Switching States in Unipolar and Hybrid Modulation

Table 2.1 summarizes the switching modes for unipolar and hybrid modulation in the H4 inverter. Unipolar modulation yields higher efficiency but higher leakage current, whereas hybrid modulation offers a balanced compromise.

Table 2.1: H4 Inverter Switching Modes: Unipolar vs Hybrid Modulation

| Unipolar Modulation |     |     |     |     |     |           | Hybrid Modulation |     |     |     |     |     |           |
|---------------------|-----|-----|-----|-----|-----|-----------|-------------------|-----|-----|-----|-----|-----|-----------|
| S1                  | S2  | S3  | S4  | D1  | D3  | $V_{out}$ | S1                | S2  | S3  | S4  | D1  | D3  | $V_{out}$ |
| On                  | Off | Off | On  | Off | Off | $V_{pv}$  | On                | Off | Off | On  | Off | Off | $V_{pv}$  |
| On                  | Off | Off | Off | Off | On  | 0         | On                | Off | Off | Off | Off | On  | 0         |
| Off                 | On  | On  | Off | Off | Off | $-V_{pv}$ | Off               | On  | On  | Off | Off | Off | $-V_{pv}$ |
| Off                 | Off | On  | Off | On  | Off | 0         | Off               | On  | On  | Off | On  | Off | 0         |

### 2.2.2 H5 Inverter Topology

The H5 inverter is a transformer-less topology designed to reduce leakage current by decoupling the PV side from the grid during freewheeling periods. Unlike the conventional H4 inverter, the H5 topology introduces an additional switch ( $Q_5$ ) at the positive DC bus, as shown in Figure 2.5. This switch, together with two clamping diodes ( $D_6$  and  $D_7$ ), helps maintain a constant common-mode voltage (CMV), thereby minimizing leakage current. The modulation strategy of H5 combines the differential mode voltage (DMV) behavior of unipolar modulation with the CMV characteristics of bipolar modulation.

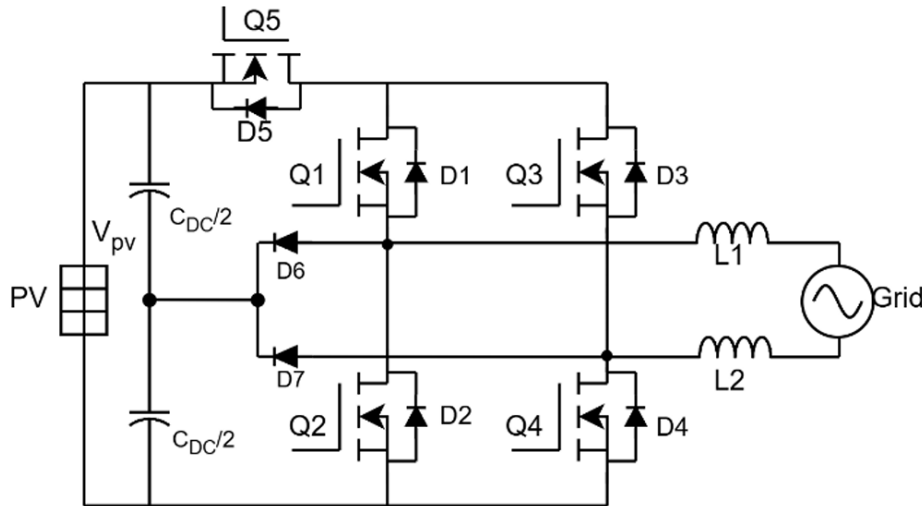


Figure 2.5: H5 inverter topology with additional switch  $Q_5$  and clamping diodes  $D_6$ ,  $D_7$ . [44]

The switching strategy of the H5 inverter is illustrated in Figure 2.6. Switches  $Q_1$  and  $Q_3$  operate at line frequency, while  $Q_2$ ,  $Q_4$ , and  $Q_5$  switch at high frequency depending on

## Types of 1 $\phi$ Transformer-less Inverter Topologies

the grid voltage polarity. This selective switching effectively reduces stress on the devices and improves efficiency.

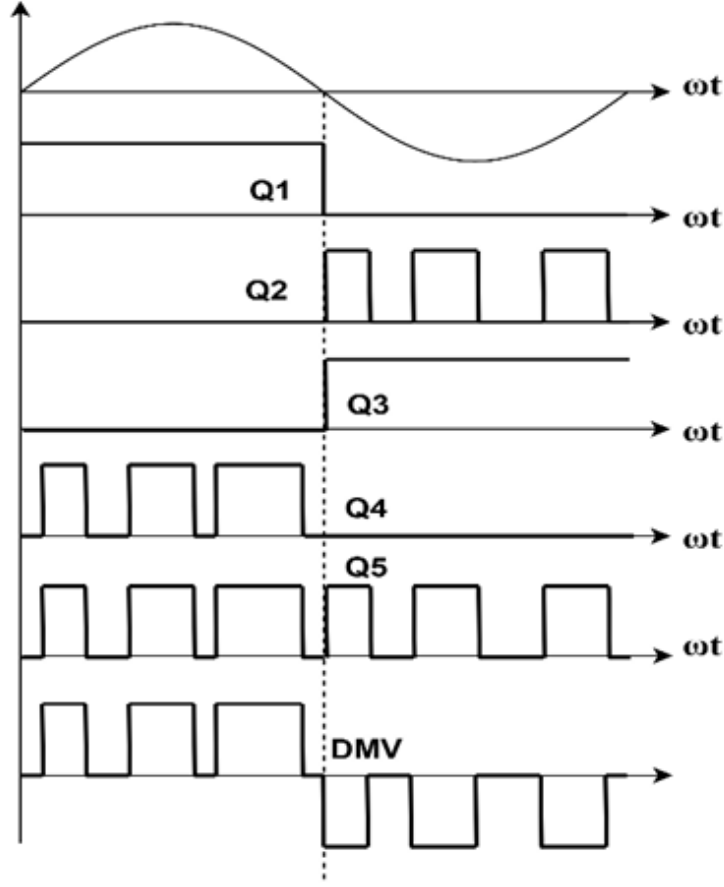


Figure 2.6: Switching pattern of H5 inverter topology [7].

### Modes of Operation in H5 Topology

The H5 inverter has four modes of operation. The extra switch  $Q_5$  and clamping diodes keep the common-mode voltage (CMV) steady, which cuts down on leakage current by a lot. Unlike H4, this configuration effectively decouples the PV side from the grid during freewheeling periods. Below are the four ways to use it. **Mode 1: Active State (Positive Half Cycle)** Switches  $Q_5$ ,  $Q_1$ , and  $Q_4$  are ON, delivering power from the PV array to the grid.

$$\text{CMV} = \frac{V_{AN} + V_{BN}}{2} = \frac{V_{pv} + 0}{2} = \frac{V_{pv}}{2}, \quad \text{DMV} = V_{pv}$$

**Mode 2: Freewheeling (Positive Half Cycle)** Only  $Q_1$  conducts while current free-wheels through  $D_3$  and the clamping diode  $D_7$ . Here, the PV side is disconnected from

the grid since  $Q_5$  is OFF.

$$\text{CMV} = \frac{V_{pv}}{2}, \quad \text{DMV} = 0$$

**Mode 3: Active State (Negative Half Cycle)** Switches  $Q_5$ ,  $Q_2$ , and  $Q_3$  are ON, transferring negative current from PV to the grid.

$$\text{CMV} = \frac{0 + V_{pv}}{2} = \frac{V_{pv}}{2}, \quad \text{DMV} = -V_{pv}$$

**Mode 4: Freewheeling (Negative Half Cycle)** Only  $Q_3$  conducts, and current free-wheels through  $D_1$  the clamping diode  $D_6$ , with the AC and DC sides completely decoupled.

$$\text{CMV} = \frac{V_{pv}}{2}, \quad \text{DMV} = 0$$

**Summary:** The H5 topology gets three-level DMV output like unipolar modulation and keeps a constant CMV like bipolar modulation. This combined benefit reduces overall harmonic distortion and leakage current, making it a very effective and dependable alternative for PV grid-connected systems that don't use transformers.

### 2.2.3 H6 Inverter Topology

The H6 inverter is an extension of the H5 topology, where an additional switch ( $S_6$ ) is inserted to reduce leakage current and conduction losses further while maintaining high efficiency. The circuit diagram of the H6 inverter is shown in Figure 2.7. The addition of  $S_6$  improves the decoupling of the PV array from the grid during freewheeling periods, ensuring a constant common-mode voltage (CMV) and suppressing leakage current.

Unlike conventional transformer-based inverters that inherently block leakage paths, transformer-less inverters directly link the PV source and the grid, which can cause leakage problems. In the H6 configuration, the CMV is maintained constant (typically between 16–20 V), effectively minimizing leakage current without requiring a transformer. The modulation is based on unipolar sinusoidal PWM, and an LCL filter is generally employed for grid interfacing.

#### Modes of Operation in H6 Topology

The H6 inverter operates in four distinct modes: two active modes (positive and negative half cycles) and two freewheeling modes. The switching states are summarized in Table 2.2.



## *Types of $1\phi$ Transformer-less Inverter Topologies*

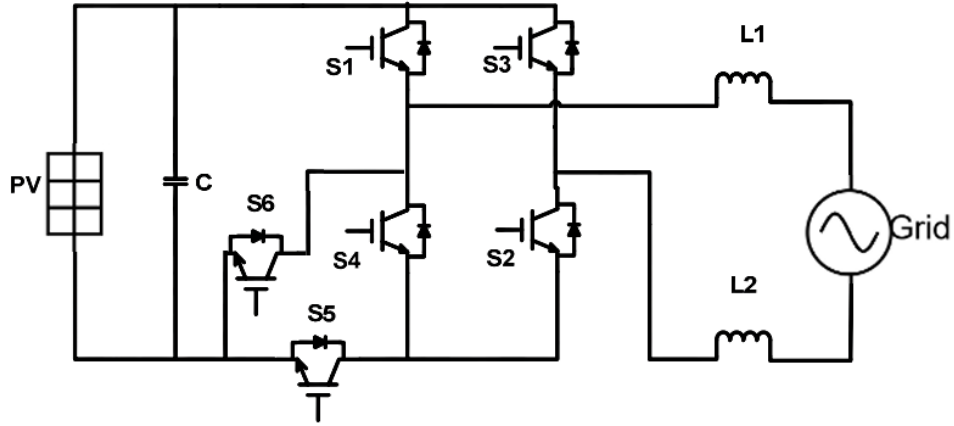


Figure 2.7: H6 Inverter Topology

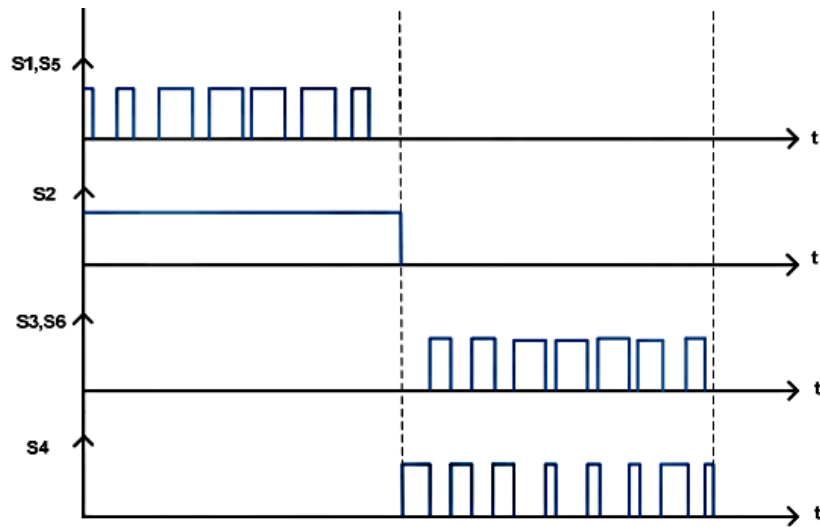


Figure 2.8: Switching pattern of H6 Inverter Topology [15].

**Mode 1: Active State (Positive Half Cycle)** Switches  $S_1$ ,  $S_4$ ,  $S_5$ , and  $S_6$  are ON, while  $S_2$  and  $S_3$  are OFF. Energy flows from the PV side to the grid.

$$\text{CMV} = \frac{V_{AN} + V_{BN}}{2} = \frac{V_{pv} + 0}{2} = \frac{V_{pv}}{2}, \quad \text{DMV} = V_{AN} - V_{BN} = V_{pv}$$

**Mode 2: Freewheeling (Positive Half Cycle)** Switch  $S_1$  conducts, and current freewheels through the antiparallel diode of  $S_3$ . The PV side is isolated since  $S_5$  and  $S_6$  are OFF.

$$\text{CMV} = \frac{V_{pv}}{2}, \quad \text{DMV} = 0$$

**Mode 3: Active State (Negative Half Cycle)** Switches  $S_2$ ,  $S_3$ ,  $S_5$ , and  $S_6$  are ON, while  $S_1$  and  $S_4$  are OFF. Negative energy is delivered from the PV side to the grid.

$$\text{CMV} = \frac{0 + V_{pv}}{2} = \frac{V_{pv}}{2}, \quad \text{DMV} = -V_{pv}$$

**Mode 4: Freewheeling (Negative Half Cycle)** Switch  $S_3$  conducts, and current flows through the antiparallel diode of  $S_1$ , while the PV side remains decoupled.

$$\text{CMV} = \frac{V_{pv}}{2}, \quad \text{DMV} = 0$$

Table 2.2: Switching states for H6 inverter topology

| S1  | S2  | S3  | S4  | S5  | S6  | D1  | D3  | $V_{out}$ |
|-----|-----|-----|-----|-----|-----|-----|-----|-----------|
| On  | Off | Off | On  | On  | On  | Off | Off | $+V_{in}$ |
| On  | Off | Off | Off | Off | Off | Off | On  | 0         |
| Off | On  | On  | Off | On  | On  | Off | Off | $-V_{in}$ |
| Off | Off | On  | Off | Off | Off | On  | Off | 0         |

**Summary:** The H6 inverter has unipolar DMV properties and keeps a constant CMV, which stops leakage current from happening. It has one more switch than H5, but its better decoupling cuts down on conduction loss and boosts overall efficiency. This makes it a good choice for PV grid applications that don't need transformers.

## 2.2.4 HERIC Inverter Topology

The Highly Efficient and Reliable Inverter Concept (HERIC) topology is a widely adopted single-phase transformerless PV inverter due to its high efficiency and excellent leakage current suppression [70]. The HERIC inverter adds two auxiliary switches ( $S_5$  and  $S_6$ ) on the AC side to decouple the PV array from the grid during freewheeling periods, thereby

## Types of 1 $\phi$ Transformer-less Inverter Topologies

maintaining a constant common-mode voltage (CMV). The circuit diagram is shown in Figure 2.9.

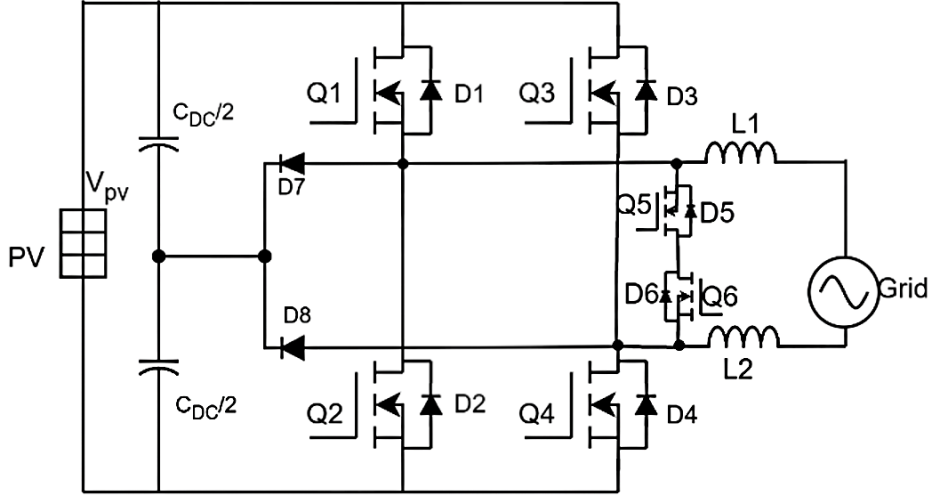


Figure 2.9: HERIC inverter topology [2].

### Modes of Operation in HERIC Topology

The HERIC inverter operates in four modes, corresponding to active and freewheeling states for both positive and negative half-cycles of the grid voltage. Each mode ensures a constant CMV, which minimizes leakage currents.

**Mode 1: Active Positive Half-Cycle** Switches  $S_1$  and  $S_4$  are on, delivering power to the grid.

$$V_{AB} = V_{dc}, \quad I_{grid} > 0, \quad CMV = \frac{V_{AN} + V_{BN}}{2} = \frac{V_{pv}}{2}$$

**Mode 2: The Freewheeling Positive Half-Cycle** Switch  $S_6$  conducts, providing a path for the grid current while the PV side is disconnected.

$$V_{AB} = 0, \quad I_{grid} > 0, \quad CMV = \frac{V_{pv}}{2}$$

**Mode 3: Active Negative Half-Cycle** Switches  $S_2$  and  $S_3$  are ON, delivering negative power to the grid.

$$V_{AB} = -V_{dc}, \quad I_{grid} < 0, \quad CMV = \frac{V_{pv}}{2}$$

**Mode 4: Freewheeling Negative Half-Cycle** Switch  $S_5$  conducts, allowing negative current to circulate while decoupling the PV array.

$$V_{AB} = 0, \quad I_{grid} < 0, \quad CMV = \frac{V_{pv}}{2}$$

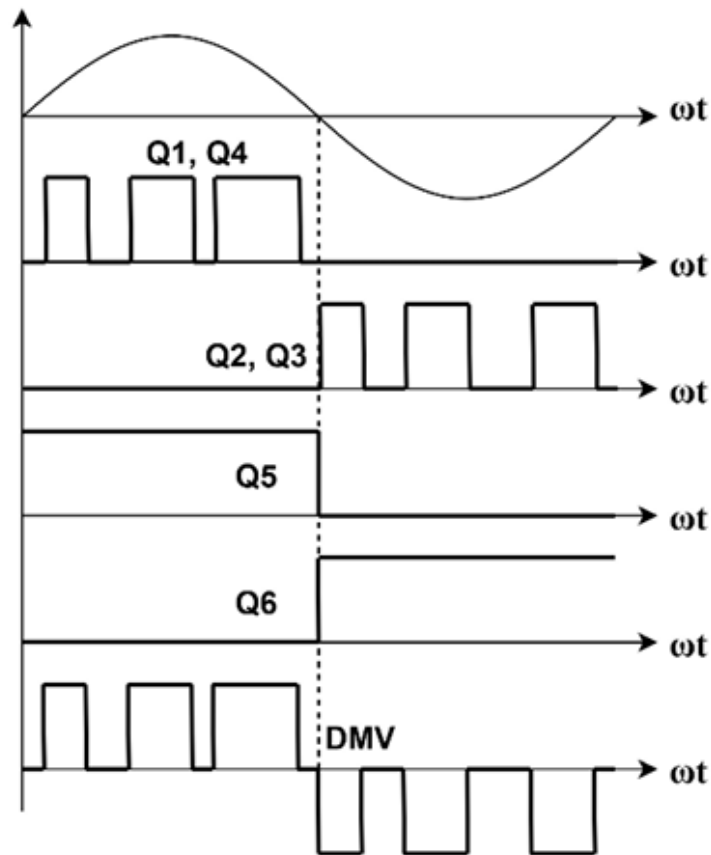


Figure 2.10: Switching pattern of HERIC inverter [52].

# Types of 1 $\phi$ Transformer-less Inverter Topologies

## Switching Characteristics

The switching states for the HERIC inverter are summarized in Table 2.3. This table shows which switches are ON for each output voltage and the corresponding grid current direction.

Table 2.3: Switching states of the HERIC inverter.

| S1  | S2  | S3  | S4  | S5  | S6  | $V_{AB}$  | $I_{grid}$ |
|-----|-----|-----|-----|-----|-----|-----------|------------|
| On  | Off | Off | On  | Off | Off | $+V_{dc}$ | $> 0$      |
| Off | Off | Off | Off | Off | On  | 0         | $> 0$      |
| Off | On  | On  | Off | Off | Off | $-V_{dc}$ | $< 0$      |
| Off | Off | Off | Off | On  | Off | 0         | $< 0$      |

**Summary:** The HERIC topology combines a simple H-bridge with two auxiliary decoupling switches, ensuring constant CMV and minimal leakage currents. Its four-mode operation maintains unipolar DMV while keeping CMV constant, reducing switching and conduction losses. This makes HERIC a highly efficient and reliable transformer-less PV inverter for grid-connected applications.

## 2.3 Comparison of Single-Phase Inverters

This section focuses on the most popular transformerless single-phase inverters—H4, H5, H6, and HERIC. The H4 inverter is designed with bipolar or unipolar PWM and is used for small PV systems. However, it is plagued with leakage current issues. With bipolar PWM, the circuit has two voltage levels ( $\pm V_{dc}$ ), which reduces leakage but increases conduction loss. Unipolar PWM has three voltage levels ( $+V_{dc}, 0, -V_{dc}$ ) and, while offering higher efficiency, results in higher leakage current. The H5 inverter adds a switch on the positive DC rail, reducing leakage current during freewheeling. However, it is slightly less efficient due to extra conduction losses. The H6 inverter, with its extra switch, improves on the H5 design by reducing leakage current and improving efficiency, making it better for high-performance systems. The HERIC inverter is the most efficient, featuring two auxiliary switches on the AC side. It has the lowest leakage current and the least loss of efficiency due to switching losses, making it the most balanced option for commercial PV inverters with high reliability and safety.

Table 2.4: Comparison of Single-Phase Transformerless Inverter Topologies

| Feature         | H4        | H5          | H6        | HERIC         |
|-----------------|-----------|-------------|-----------|---------------|
| Switches        | 4         | 5           | 6         | 6 (AC bypass) |
| Leakage Current | High      | Low         | Very Low  | Very Low      |
| Efficiency      | Medium    | Medium-High | High      | Very High     |
| Complexity      | Low       | Medium      | Higher    | Medium        |
| Remarks         | ↑ Leakage | PV isolated | Efficient | ↓leakage      |

## 2.4 Critical Analysis and Comparison with Recent Works

The comparison analysis indicates that increasing the number of switches and implementing proper decoupling techniques gradually reduces leakage current as it transitions from H4 to H5 and then to H6/HERIC. The H4 architecture is the easiest and cheapest to use, although it has larger leakage currents, especially when unipolar modulation is used. Recent studies, like the research by Kołodziejewski et al. (2022) [60] and Yue et al. (2020) [1], have emphasized the benefits of H4 regarding its simplicity and inexpensive cost, while also addressing the considerable leakage current challenges linked to this topology. Yue et al. (2020) [1] suggested ways to reduce leakage currents; however, these methods are still not as beneficial as more complicated topologies like H5 and H6. The H4 inverter is still extensively used in smaller PV systems where cost is a big deal, even if it has several problems. However, it doesn't work as well when leakage suppression is the most important thing.

The H5 topology adds a switch and clamping diodes to separate the PV side from the grid during freewheeling periods. This technique cuts down on leakage currents. But this makes things more complicated and causes conduction losses. Phuyal et al. (2024) [6] say that the H5 inverter does a better job of stopping leakage current, but it is less efficient than more sophisticated topologies like H6 since it loses more power through conduction. Shrestha et al. (2025) [13] conducted a more in-depth analysis of the balance between efficiency and cost, demonstrating that the H5 inverter's performance enhancements would not warrant the additional cost for residential PV systems. Still, the H5 inverter is better at lowering leakage current than the H4 inverter, which makes it suitable for medium-sized projects.

The H6 inverter is an improvement on the H5 arrangement because it has a sixth switch that separates the PV source from the grid even more. This makes it more efficient and reduces leakage current. The H6 inverter works better than the H5 inverter, especially in large-scale PV systems, because it reduces leakage currents and conduction losses (Zhang et al., 2021 [10] and Rizzoli et al., 2016 [7]). But the H6 topology is less

## *Types of 1 $\phi$ Transformer-less Inverter Topologies*

cost-effective and tougher to use in smaller applications since it adds to the complexity of control and system design. The H6 inverter is still thought to be a very efficient choice for large PV systems where performance is the most important thing.

The HERIC topology is the best of the transformerless inverters because it keeps a constant common-mode voltage (CMV), which cuts down on leakage currents while keeping high efficiency. Shrestha et al. (2025) [13] and Guo et al. (2025) [4] discussed how HERIC can help reduce leakage current and make systems more reliable. HERIC works very well in both homes and businesses, unlike the H4 and H5 topologies. But adding more auxiliary switches complicates the system, which boosts the cost and makes it harder to design and build. Phuyal et al. (2024) [6] and Tang et al. (2019) [22] said that the HERIC inverter is very efficient, but its complexity improves it for big projects.

Recent research, including the comparative analysis by Shrestha et al. (2025) [5], has shown that the HERIC topology has the lowest leakage current and the maximum efficiency, although it is more complicated and expensive than H4 and H5. Li et al. (2020) [12] and Tang et al. (2019) [22] have also talked about how employing sophisticated modulation techniques can improve HERIC work by making it more efficient and lowering the leakage current. HERIC is now a dependable and efficient option for high-performance PV systems thanks to these enhancements.

To sum up, each of the H4, H5, H6, and HERIC inverter topologies has its pros and cons. The H4 inverter is the easiest to use and the cheapest, but it doesn't work well for bigger systems since it doesn't stop leakage current. The H5 inverter does a better job of stopping leakage current, but it is more complicated and has more conduction losses. The H6 inverter is an even superior version of the H5. It has higher isolation and less conduction loss, which makes it perfect for high-performance applications. The HERIC topology is the greatest choice for large-scale commercial uses because it is the most efficient, prevents leakage current, and is the most reliable. Future research, as articulated by Kołodziejewski et al. (2022) [69] and Shrestha et al. (2025) [5], must persist in emphasizing the simplification of control mechanisms and the reduction of costs associated with these topologies, all while preserving their elevated efficiency and safety standards.

# Chapter 3

## Switching Devices and Techniques

### 3.1 Introduction

Selecting the appropriate switching components and techniques is crucial for single-phase transformer-less inverter design to ensure performance, efficiency, and suitability for a given application. Two switching devices that are frequently utilized in modern inverter architecture are the Insulated Gate Bipolar Transistor (IGBT) and the Metal-Oxide-Semiconductor Field-Effect Transistor (MOSFET). The stability and quality of the inverter waveform are also significantly impacted by switching techniques such as pulse width modulation (PWM) and PID-based control.

### 3.2 Switching Devices

#### 3.2.1 MOSFETs (Metal-Oxide-Semiconductor Field-Effect Transistors)

A MOSFET is a three-terminal semiconductor device (Gate, Drain, Source) that can be used to switch or amplify signals. MOSFETs are commonly used in low- to medium-power inverters and high-frequency applications because they have high input impedance and low conduction losses.

#### Specifications:

- Voltage rating: Up to 600–1000 V



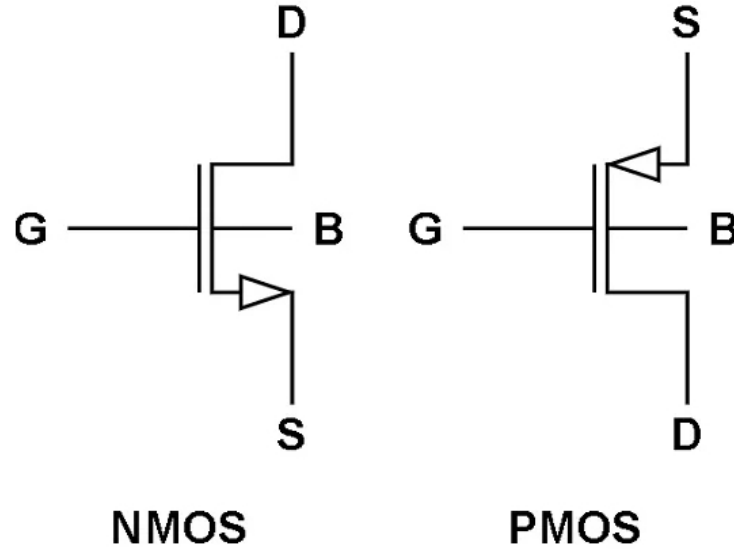


Figure 3.1: Symbol of a MOSFET [19].

- Switching frequency: Up to 100 kHz
- On-state resistance: Very low  $R_{DS(on)}$
- Best suited for: Low/medium voltage inverters, DC-DC converters

Table 3.1: Advantages and Disadvantages

| Advantages                                              | Disadvantages                                    |
|---------------------------------------------------------|--------------------------------------------------|
| 1. Very fast switching speed (high-frequency operation) | 1. Limited voltage handling (usually below 1 kV) |
| 2. Low gate drive power requirement                     | 2. Conduction losses increase at higher voltages |
| 3. Excellent efficiency in low-voltage ranges           |                                                  |

### 3.2.2 IGBTs (Insulated Gate Bipolar Transistors)

An IGBT combines the MOSFET input structure with the capabilities of a bipolar transistor to handle current. IGBTs are a popular choice for renewable energy inverters, motor drives, and electric vehicles (EVs) since they can manage high power efficiently because they are both types of devices.

#### Specifications:

- Voltage rating: Up to several kV
- Switching frequency: 1–20 kHz
- Current handling: Very high

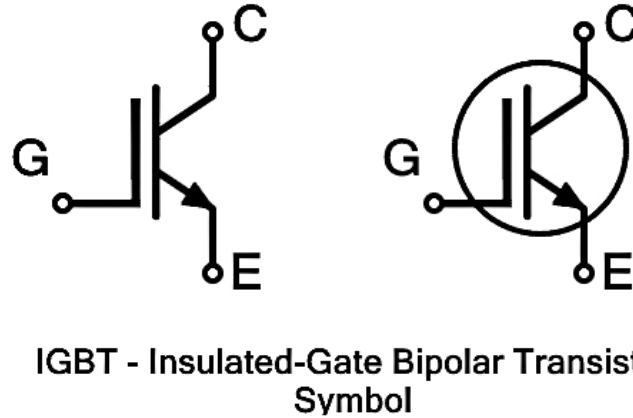


Figure 3.2: Symbol of an IGBT [19].

- Best suited for: Medium/high power inverters, motor drives, renewable energy systems

Table 3.2: Advantages and Disadvantages

| Advantages                                            | Disadvantages                           |
|-------------------------------------------------------|-----------------------------------------|
| 1. Handles very high voltages efficiently             | 1. Slower switching than MOSFETs        |
| 2. Lower conduction losses at high voltages           | 2. Higher gate drive power requirements |
| 3. Strong performance in medium switching frequencies |                                         |

### 3.3 Comparison of MOSFETs and IGBTs

Table 3.3: Comparison between MOSFETs and IGBTs

| Feature            | MOSFET                              | IGBT                               |
|--------------------|-------------------------------------|------------------------------------|
| Switching speed    | Very high (tens to hundreds of kHz) | Moderate (1–20 kHz)                |
| Voltage range      | Up to 1 kV                          | Up to several kV                   |
| Conduction losses  | Low at low voltages                 | Low at high voltages               |
| Gate drive         | Simple, low power                   | Complex, higher power              |
| Application domain | Low/medium power, high frequency    | Medium/high power, lower frequency |

### 3.4 Switching Techniques

In transformerless inverters, switching techniques play a crucial role in controlling the output voltage and minimizing the leakage current while maintaining efficiency. Some commonly used switching techniques are:

#### 3.4.1 Pulse Width Modulation (PWM)

Pulse Width Modulation (PWM) is the most widely used technique in inverters, where the width of the pulses is modulated to control the effective output voltage and reduce harmonics. By varying the width of the pulses while keeping the frequency constant, PWM controls the amount of power delivered to an electrical device. The effective output voltage is determined by the duty cycle of the PWM signal, and it can be represented as a phasor with an amplitude that corresponds to the duty cycle and a phase that aligns with the reference waveform.[19]

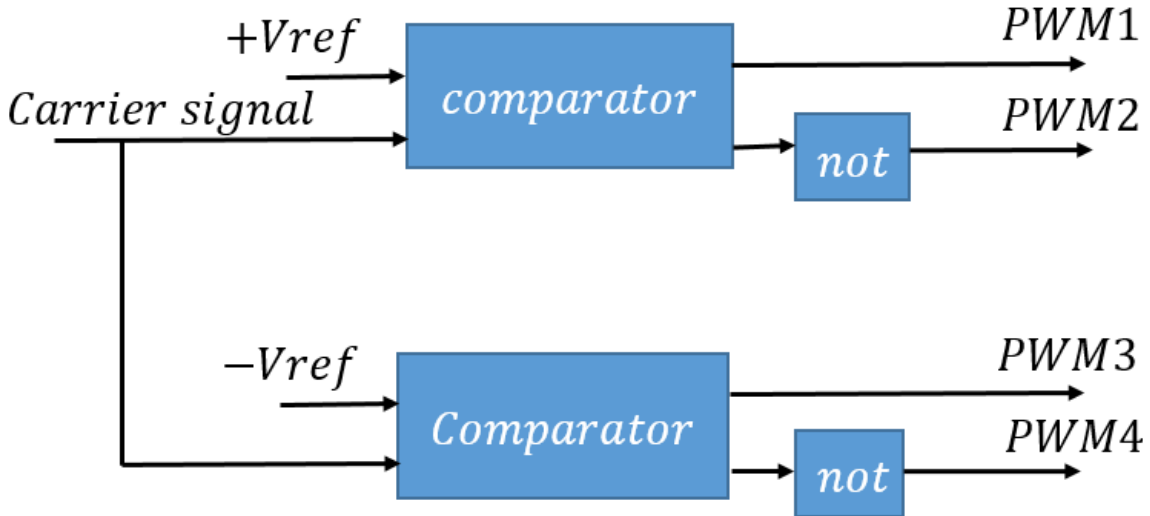


Figure 3.3: Unipolar PWM Generator Block

#### 3.4.2 Sinusoidal PWM (SPWM)

The Sinusoidal Pulse Width Modulation (SPWM) method utilizes a sinusoidal modulating signal compared to a triangular carrier signal to produce switching pulses, which results in an output waveform that closely resembles an ideal AC sine wave. The modulating signal, shown in Figure 3.5, is compared to the carrier signal to generate the switching signals  $S_1$  and  $S_2$ , which control the operation of the inverter.



Figure 3.4: Bipolar PWM Generator Block

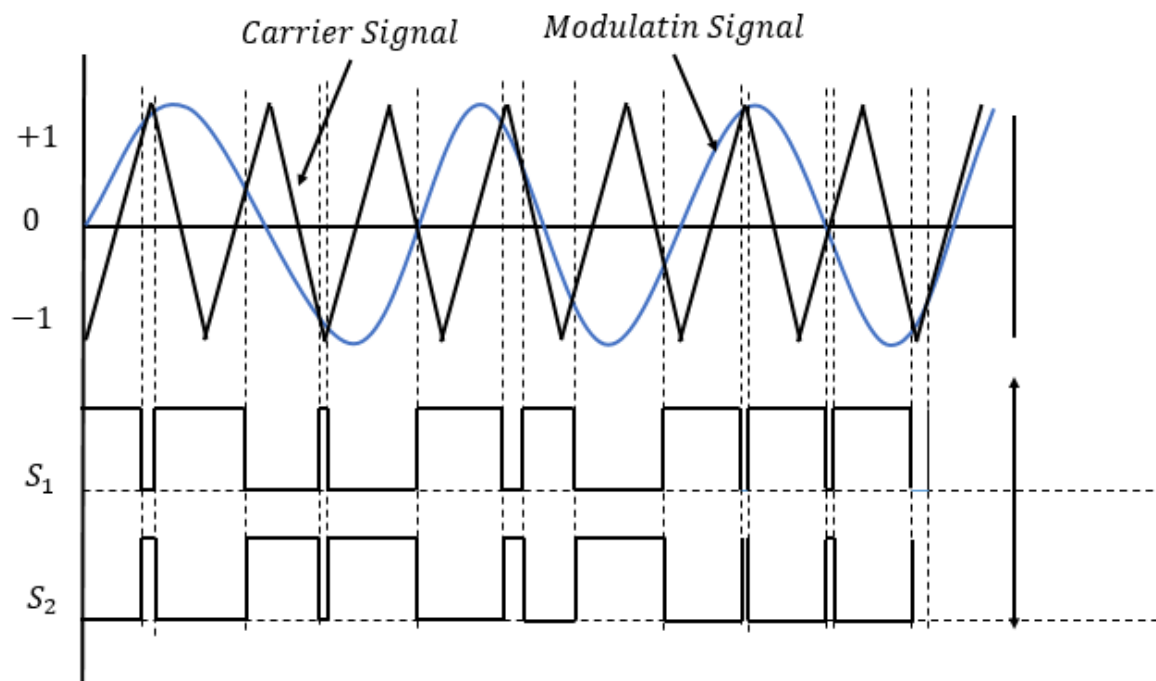


Figure 3.5: SPWM Modulating Signal with Carrier.

### 3.4.3 PID-based Control

PID controllers are frequently combined with PWM for regulation of output voltage and current in inverters. The Proportional, Integral, and Derivative terms improve stability, dynamic response, and error minimization under load variations.

#### Controller Overview

In a closed-loop system, a controller plays a crucial role in obtaining the desired output in real time. It is based on a feedback mechanism that continuously monitors the system output, compares it to the desired set value, and adjusts its control signal to minimize the error. Essentially, a control system is a mechanical or electronic device that automatically regulates a system to maintain a desired state without any human operator.

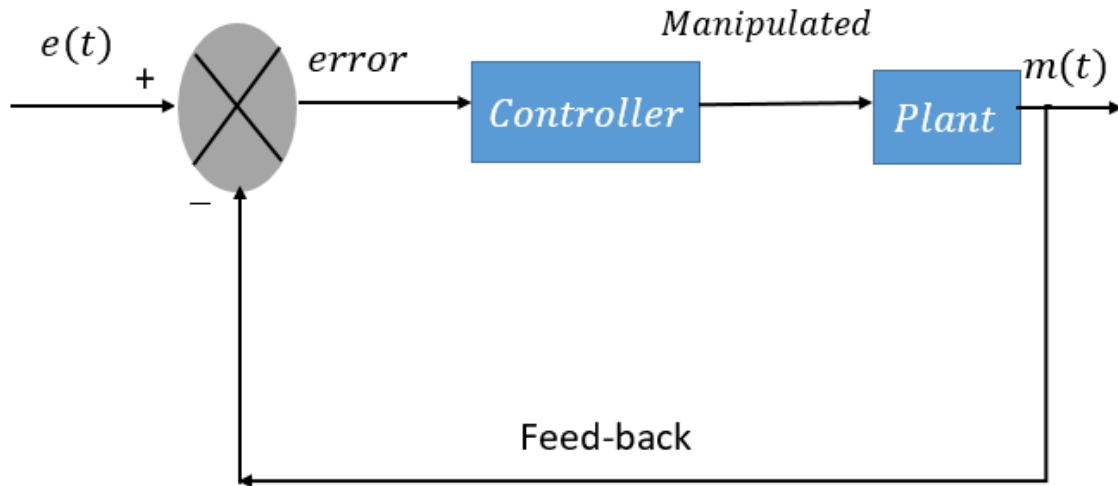


Figure 3.6: A Closed Loop System

Based on the combination of control actions, the following types of controller are commonly used:

#### Proportional (P) Controller

The output of a proportional controller is directly proportional to its input:

$$m(t) = K_p e(t)$$

By Laplace transform:

$$m(s) = K_p e(s)$$

For voltage controller:  $K_p = \frac{\text{filter capacitance}}{\text{time constants}}$

For current controller:  $K_p = \frac{\text{filter inductance}}{\text{time constants}}$  The proportional controller provides a fast response, but it considers only the present error and cannot eliminate steady-state error.



Figure 3.7: Block Diagram of P Controller

### Proportional-Integral (PI) Controller

The PI controller combines proportional and integral action:

$$m(t) = K_p e(t) + K_i \int e(t) dt$$

Laplace transform:

$$m(s) = e(s) \left( K_p + \frac{K_i}{s} \right)$$

For voltage controller:  $K_p = \frac{\text{filter capacitance}}{\text{time constants}}$ ,  $K_i = \frac{\text{capacitor ESR}}{\text{time constants}}$

For the current controller:  $K_p = \frac{\text{filter inductance}}{\text{time constants}}$ ,  $K_i = \frac{\text{series resistance}}{\text{time constants}}$

The PI controller eliminates steady-state error while maintaining a stable response.

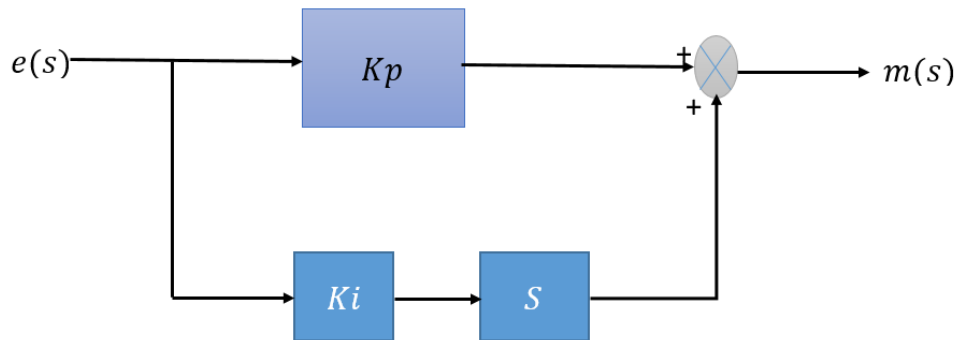


Figure 3.8: Block Diagram of PI Controller

### Proportional-Derivative (PD) Controller

The PD controller combines proportional and derivative actions:

$$m(t) = K_p e(t) + K_d \frac{d}{dt} e(t)$$

Laplace transform:

$$m(s) = e(s)(K_p + K_d s)$$

For voltage controller:  $K_p = \frac{\text{filter capacitance}}{\text{time constants}}$ ,  $K_d = \frac{\text{filter capacitance} \times \text{series resistance}}{\text{time constants}}$   
 For current controller:  $K_p = \frac{\text{filter inductance}}{\text{time constants}}$ ,  $K_d = \frac{\text{filter inductance} \times \text{series resistance}}{\text{time constants}}$  PD controllers improve transient response and reduce overshoot but cannot eliminate steady-state error.

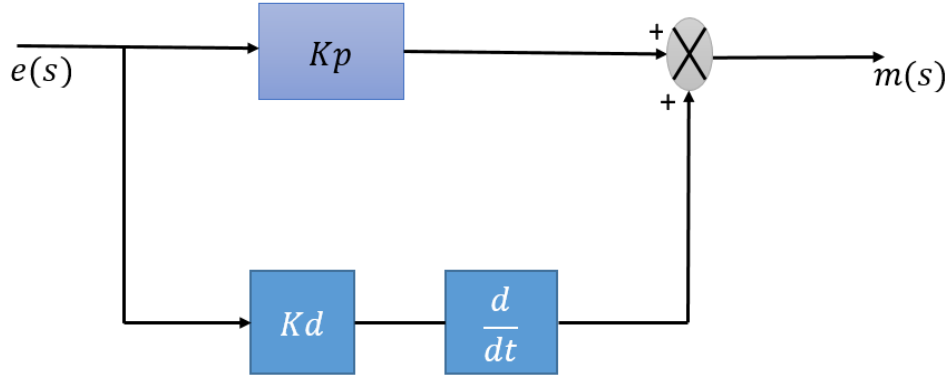


Figure 3.9: Block Diagram of PD Controller

### Proportional-Integral-Derivative (PID) Controller

The PID controller combines proportional, integral, and derivative actions:

$$m(t) = K_p e(t) + K_d \frac{d}{dt} e(t) + K_i \int e(t) dt$$

Laplace transform:

$$m(s) = e(s) \left( K_p + K_d s + \frac{K_i}{s} \right)$$

For voltage controller:  $K_p = \frac{\text{filter capacitance}}{\text{time constants}}$ ,  $K_d = \text{time constants} \times \text{capacitance}$ ,  $K_i = \frac{\text{ESR}}{\text{time constants}}$   
 For current controller:  $K_p = \frac{\text{filter inductance}}{\text{time constants}}$ ,  $K_d = \text{time constants} \times \text{inductance}$ ,  $K_i =$

series resistance  
time constants In transformer-less grid-connected inverters, PID controllers provide balanced, fast, and precise regulation of voltage and current by eliminating steady-state error and improving transient response.

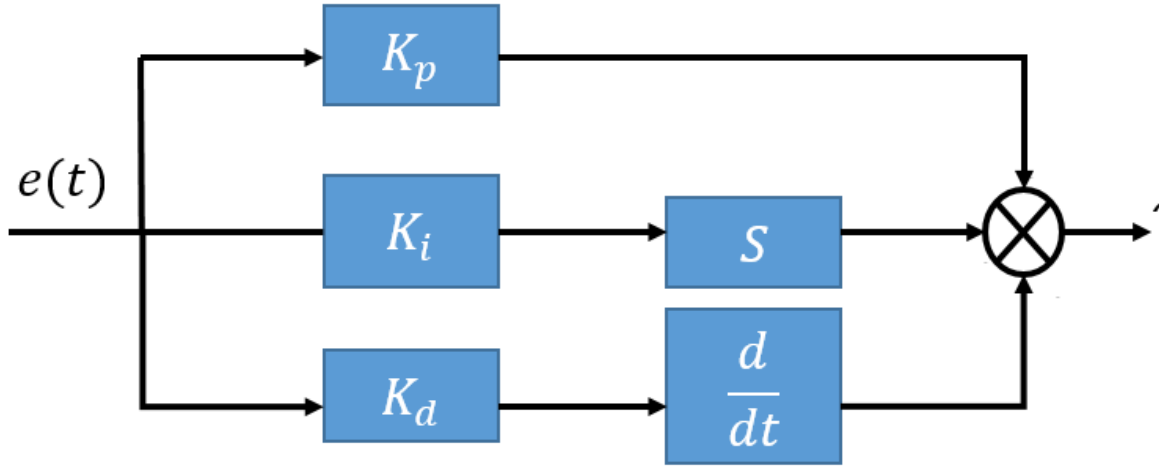


Figure 3.10: Block Diagram of PID Controller

### 3.4.4 Other Techniques

- Space Vector PWM (SVPWM) — superior harmonic performance and higher DC bus utilization.
- Hysteresis control — simple implementation with fast response but variable switching frequency.
- Model Predictive Control (MPC) — advanced control offering optimal dynamic performance in modern high-efficiency systems.

## 3.5 Conclusion

The application domain significantly affects the choice between MOSFETs and IGBTs. IGBTs are better for applications that need a lot of power and voltage, while MOSFETs are better for applications that need a lot of frequency and low to medium power. Also, advanced switching techniques like PWM and PID-based regulation make sure that the inverter works better, is more efficient, and has better waveform quality.



# Chapter 4

## LC Filter

### 4.1 Introduction

A filter is an electrical network used to alter a signal's frequency that is composed of components such as resistors (R), capacitors (C), and inductors (L). AC input is converted to pure DC by an inverter. The pulsing DC output contains both AC and DC components. A DC input is converted to pure AC by an inverter. Its output is AC that pulses and contains harmonics. Without filters, the grid-connected inverter's current contains a lot of harmonics. A number of power quality issues arise when harmonics are introduced to the current and sent into the grid, worsening the grid voltage. Connected devices are unable to function properly because they require pure AC.

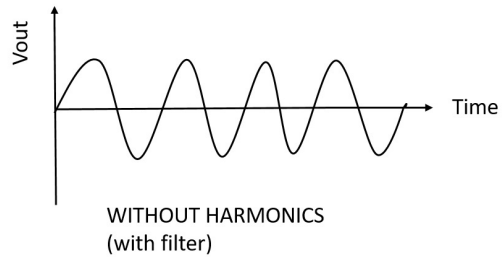
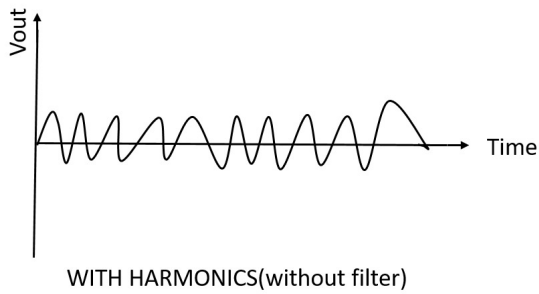


Figure 4.1: Before filter with harmonics [38].

Figure 4.2: After filter without harmonics

In transformer-less on-grid inverters, the switching action of semiconductor devices (IGBT or MOSFET) generates harmonics in the output voltage and current. If these harmonics are directly fed into the grid, they can cause power quality issues, increased losses, electromagnetic interference (EMI), and even malfunction of sensitive devices such as

MRI machines, ECG monitors, and programmable logic controllers. Therefore, filters are used at the inverter output to ensure compliance with standards such as IEEE 519 and IEC 61000. Filters suppress switching frequency harmonics and provide a near-sinusoidal output by allowing the fundamental frequency to pass while attenuating all other frequencies. This ensures that the total harmonic distortion (THD) remains within the allowable limits for connection to the grid. According to the IEEE 519 and IEC 61000 standards, the THD limits for both current and voltage must be  $\leq 5\%$  (often stricter at  $\leq 3\%$ ). Another advantage of filters is that they provide protection for both the inverter and the grid from high-frequency stresses, while also improving overall efficiency and power quality.

## 4.2 Types of Filters

Several types of filters can be used at the inverter output to obtain a near-sinusoidal waveform.

### 4.2.1 L Filter

An L filter is the simplest type of output filter, consisting of a single inductor in series with the inverter output. Its primary function is to suppress high-frequency switching harmonics while allowing the fundamental frequency (50 Hz in Bangladesh) to pass.

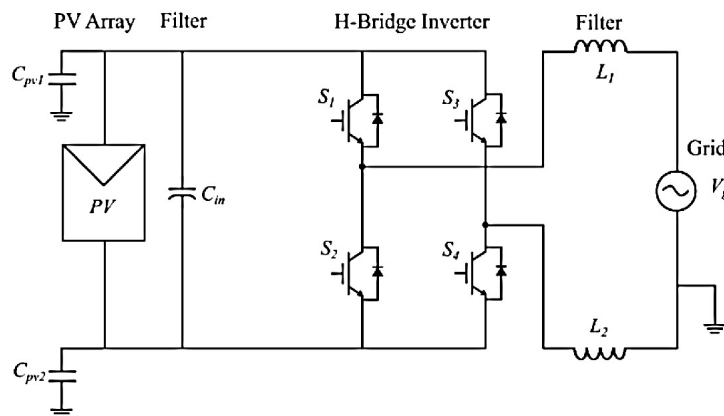


Figure 4.3: A low-pass L filter.

current, Leakage current depends on common mode voltage, which depends on switching frequency. If common mode voltage increases, then the corresponding leakage current increases and vice versa. In the case of the LC filter and LCL filter, there is a capacitor used on the grid side, which could increase common mode voltage; as a result, leakage

current also increases. But in an L filter there is no capacitor used on the grid side; as a result, the issue of common mode voltage doesn't arise.

Since the L filter only used an inductor, there are lots of advantages over the LC filter and LCL filter, like reduced cost, simple design, low power loss, and small size. Due to those advantages, it is an efficient choice for prototype transformer-less on-grid systems. Its ability to provide adequate harmonic attenuation while maintaining system safety and simplicity makes it the preferred filter for low- to medium-power transformerless inverters.

These advantages make the L filter an efficient choice for low- to medium-power transformerless on-grid PV systems.

### 4.2.2 LC Filter

An LCL filter is a high-output-type low-pass filter among different types of filters used in grid-connected transformerless inverters, consisting of two series inductors (L), one on the inverter side and one on the grid side, with a capacitor (C) connected between them. This combination of two inductors and a capacitor provides better harmonic attenuation compared to a simple L filter and LC filter. The basic function is very simple, which is when the output of the inverter is fed to the filter, it suppresses high-frequency switching harmonics and allows a fundamental frequency (which is 50 Hz in Bangladesh) to pass; therefore, a pure sinusoidal output is obtained to the grid.

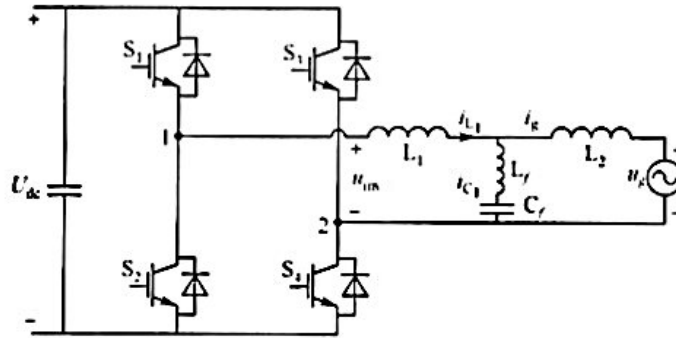


Figure 4.4: A low-pass LC filter.

The capacitor helps bypass high-frequency components, while the inductor limits the current ripple generated by the rapid switching of power semiconductor devices (IGBT or MOSFET) as a result of better power quality and effectively reduces both voltage and current Total Harmonic Distortion (THD). An LC filter provides better harmonic attenuation compared to a simple L filter, making it very suitable for systems that must comply with strict grid standards such as IEEE 519 and IEC 61000.

In the case of leakage current, an LC filter helps reduce this problem by attenuating high-frequency switching components, which are the primary contributor to the leakage. But the capacitor used in the LC filter provides a path for some leakage currents. Therefore, in transformer-less inverter designs, the values of the inductor and capacitor have to be chosen so as to achieve harmonic reduction without significantly increasing leakage current.

### 4.2.3 LCL Filter

An LCL filter is a high-output-type low-pass filter among different types of filters used in grid-connected transformerless inverters, consisting of two series inductors (L), one on the inverter side and one on the grid side, with a capacitor (C) connected between them. This combination of two inductors and a capacitor provides better harmonic attenuation compared to a simple L filter and LC filter. The basic function of the filter is straightforward: when the inverter's output is fed into it, the filter suppresses high-frequency switching harmonics and allows a fundamental frequency of 50 Hz (as in Bangladesh) to pass, resulting in a pure sinusoidal output to the grid.

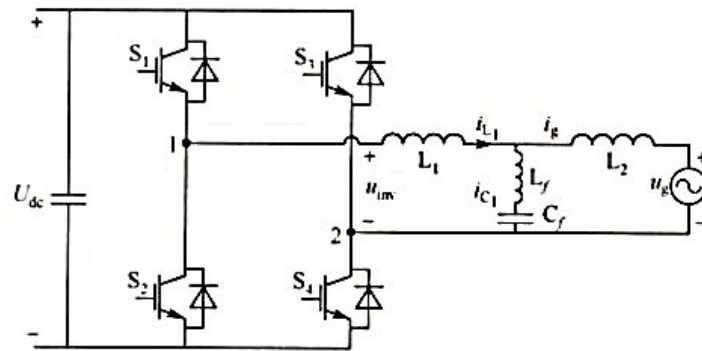


Figure 4.5: A basic LCL filter.

The inverter-side inductor (L1) limits high-frequency switching ripple and reduces the current changes, also mitigating leakage current. The grid-side inductor (L2) smooths the current injected into the grid. The shunt capacitor bypasses high-frequency voltage components, further improving harmonic suppression. But in the case of a transformer-less inverter, this shunt capacitor provides a path for some leakage current. So in the case of LCL filters in transformer-less inverter designs, the values of the inductor and capacitor have to be chosen so as to achieve harmonic reduction without significantly increasing leakage current.

#### An Basic LCL Filter Design Formula

##### 1. Design of Inductor

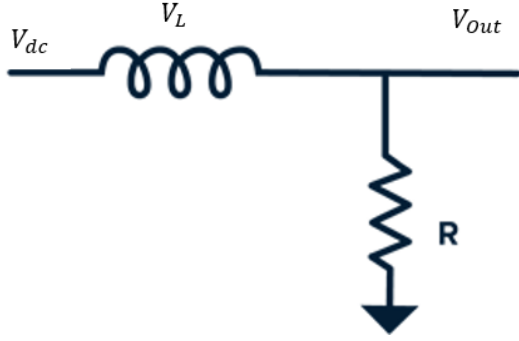


Figure 4.6: A basic L filter.

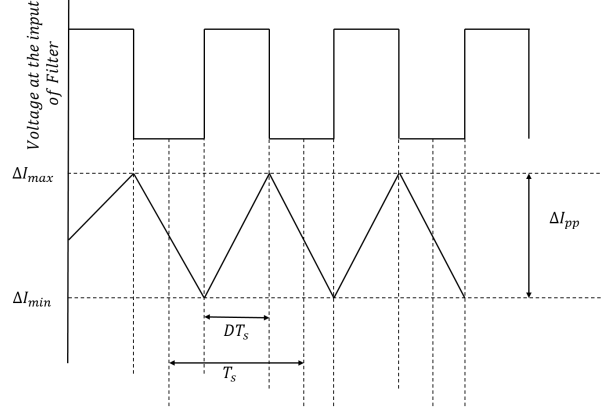


Figure 4.7: Input of Filter.

The inductor ripple current  $\Delta I_{pp}$  is defined as:

$$\Delta I_{pp} = \Delta I_{\max} - \Delta I_{\min} \quad (4.1)$$

From the inductor voltage equation:

$$V_L = L_i \frac{di}{dt} \quad (4.2)$$

For a PWM inverter:

$$V_{dc} - V_{out} = L_i \frac{di}{dt} \quad (4.3)$$

$$V_{dc} - V_{out} = L_i \frac{\Delta I_{pp}}{DT_s} \quad (4.4)$$

$$\Delta I_{pp} = \frac{DT_s(V_{dc} - V_{out})}{L_i} \quad (4.5)$$

At any time, for a PWM inverter with duty cycle  $D$ :

$$D(\omega t) = m_a \sin(\omega t), \quad \text{where } m_a \text{ is the modulation index.} \quad (4.6)$$

The output voltage is:

$$V_{out} = V_{dc} \cdot D \quad (4.7)$$

Substituting  $V_{\text{out}}$  in equation (4.5):

$$\Delta I_{pp} = \frac{m_a \sin(\omega t) T_s (V_{\text{dc}} - V_{\text{dc}} m_a \sin(\omega t))}{L_i} \quad (4.8)$$

$$\Delta I_{pp} = \frac{m_a \sin(\omega t) T_s V_{\text{dc}} (1 - m_a \sin(\omega t))}{L_i} \quad (4.9)$$

To find the modulation index corresponding to the **maximum ripple**, set the derivative to zero:

$$\frac{d}{dt}(\Delta I_{pp}) = 0 \quad \Rightarrow \quad \sin(\omega t) = \frac{1}{2m_a} \quad (4.10)$$

The maximum ripple current is then:

$$\Delta I_{pp,\text{max}} = \frac{V_{\text{dc}} T_s}{4L_i} \quad (4.11)$$

Finally, the required inductance to the Inverter Side:

$$L_1 = \frac{V_{\text{dc}}}{4F_{\text{SW}} \Delta I_{pp,\text{max}}} \quad (4.12)$$

2. Design of Inductor L2: The total Inductance ( $L_1 + L_2$ ) is selected based on the maximum voltage drop across the inductor. Maximum voltage drop is limited to 10% of the rated voltage.

$$V_{L_1+L_2} = I \cdot X(L_1 + L_2) = I \cdot 2\pi f (L_1 + L_2) \quad (4.13)$$

$$I \cdot 2\pi f (L_1 + L_2) = 10\% \text{ of } V$$

$$L_1 + L_2 = \frac{10\% \text{ of } V}{I \cdot 2\pi f}$$

$$I = \frac{S}{V}$$

$$\begin{aligned} L_1 + L_2 &= \frac{0.1 V}{\frac{S}{V} \cdot 2\pi f} \\ &= \frac{0.1 V^2}{2\pi f S} \end{aligned}$$

Finally, the required inductance to the grid side:

$$L_2 = \frac{0.1 V^2}{2\pi f S} - L_1 \quad (4.14)$$

3. Design of Capacitor: Capacitor is designed based on the reactive power at the rated conditions. Reactive power  $Q$  absorbed by the capacitor is limited to 5% of the rated power  $S$ :

$$Q = V^2 \cdot \frac{1}{2\pi f C}$$

$$V^2 \cdot \frac{1}{2\pi f C} = 0.05 S \quad (5\% \text{ of } S)$$

$$2\pi f C \cdot V^2 = 0.05 S$$

Finally, the required capacitance:

$$C = \frac{0.05 S}{2\pi f V^2} \quad (4.15)$$

Now, finding the resonant frequency:

$$f_{\text{res}} = \frac{1}{2\pi} \cdot \sqrt{\frac{L_1 + L_2}{L_1 \cdot L_2 \cdot C}} \quad (4.16)$$

For proper working of the filter:

$$10f_g < f_{\text{res}} < 0.5f_s \quad (4.17)$$

$$f_g = \text{grid frequency} \quad (4.18)$$

$$f_{\text{res}} = \text{resonant frequency} \quad (4.19)$$

$$f_s = \text{switching frequency} \quad (4.20)$$

### 4.3 Comparative Analysis of Filters

A comparative analysis between L, LC, and LCL filters is presented in Table 4.1.

Table 4.1: Comparison between L, LC, and LCL Filters

| Criteria                    | L Filter                   | LC Filter                   | LCL Filter                             |
|-----------------------------|----------------------------|-----------------------------|----------------------------------------|
| <b>Complexity</b>           | Very Low                   | Moderate                    | High (requires careful design/damping) |
| <b>Cost</b>                 | Low                        | Medium                      | High (more components and control)     |
| <b>Harmonic Attenuation</b> | Low                        | Moderate (2nd order)        | High (3rd order)                       |
| <b>Efficiency</b>           | High (low component count) | Good (some reactive losses) | Lower (more losses due to components)  |
| <b>Stability</b>            | Very Stable                | Risk of Resonance           | Needs damping (active or passive)      |

### 4.4 Conclusions

Between the L, LC, and LCL filters, the L filter was chosen for our work because, in the case of the bipolar HERIC topology, it provided the best output performance. Its ability to provide adequate harmonic attenuation while avoiding additional leakage paths makes it the most suitable option for our prototype grid-connected transformerless inverter system.



# Chapter 5

## Analysis of Simulation Results

This chapter uses simulations to look at different inverter topologies in a PV on-grid system to determine ways to lower leakage current. The main goal is to prove the theoretical ideas from earlier chapters by showing and analyzing the simulation results from MATLAB Simulink. To come to a conclusion about which method is the most efficient, the results from different topologies and pulse width modulation (PWM) techniques will be compared.

### 5.1 Simulation Setup and Methodology

The simulation scenario and setting played a critical role in our study.

We have chosen MATLAB Simulink for the complete implementation of the project, as it is software that can model power electronic systems within maximum detail and is proven to be accurate in academic and industrial research. We further enriched the experimental environment by defining the simulation parameters as standard. Actually, we, the authors, are lacking this sort of benchmark to guarantee any differences observed, for example, in efficiency or in leakage current, could indeed be attributed to the inverter topology we were comparing, instead of modifications in the environment we had applied a prior. general simulation

Table 5.1: Basic simulation parameters

| Parameter                 | Value / Notes                             |
|---------------------------|-------------------------------------------|
| Software                  | MATLAB Simulink (Electronics toolbox)     |
| Input PV voltage          | 220 V, 460 V (depending on topology)      |
| Grid frequency            | 50 Hz                                     |
| Switching frequency range | 5 kHz – 40 kHz (varies per experiment)    |
| Solver                    | Continuous (Powergui)                     |
| PWM types                 | Unipolar SPWM, Bipolar SPWM               |
| Power devices             | MOSFET / IGBT (as specified per topology) |

## 5.2 Analysis of Grid-Tied Topologies

This section shows what happens with each type of grid-tied inverter. For each topology, we give you a simulation model (a placeholder), a table of parameters, placeholders for result figures, and a short prompt for analysis.

### 5.2.1 H4 Unipolar Topology

This part looks at the H4 unipolar inverter topology, which was chosen because it is easy to use and can efficiently minimize leakage current while keeping efficiency high. There is a full analysis of the simulation parameters, the MATLAB results, and what they mean.

Table 5.2: H4 Unipolar: Simulation Parameters

| Parameter                        | Value / Notes                    |
|----------------------------------|----------------------------------|
| DC link capacitance ( $C_{dc}$ ) | 0.94 F                           |
| Series resistance ( $R_{dc}$ )   | 0.049 $\Omega$                   |
| Input DC voltage ( $V_{dc}$ )    | 220 V                            |
| Switching frequency ( $f_s$ )    | e.g. 1 kHz – 25 kHz (test sweep) |
| Filter inductance ( $L1, L2$ )   | 0.09 H                           |
| Parasitic capacitance            | $150 \times 10^{-9}$ F           |
| Load resistance ( $R_L$ )        | 50 $\Omega$                      |
| Simulation stop time             | 0.08 s                           |
| PWM type                         | Unipolar SPWM                    |
| Power device                     | MOSFET                           |
| Solver                           | Continuous (Powergui)            |

The DC link capacitance of 0.94 F and the parasitic capacitance of  $150 \times 10^{-9}$  F were selected based on standardized sizing methods that relate capacitance to panel power rating (nF/kW to mF/kW) and environmental conditions such as humidity and rain [8],[1]. Crystalline PV modules typically require 50–150 nF/kW, while thin-film modules

## Analysis of Simulation Results

can require up to 1 mF/kW. Wet surfaces can dramatically increase capacitance due to water's high relative permittivity. The series resistance value of  $0.049\ \Omega$  was chosen based on typical resistance values found in PV systems, especially for grid-tied inverters [7],[6]. This resistance is influenced by factors such as the length and material of the wiring and the overall system layout. While lower resistance values would reduce power loss, they would also increase system cost, which is why a moderate value was selected for this simulation. These parameters ensure the simulation reflects realistic operating conditions, impacting efficiency and the overall performance of the inverter.

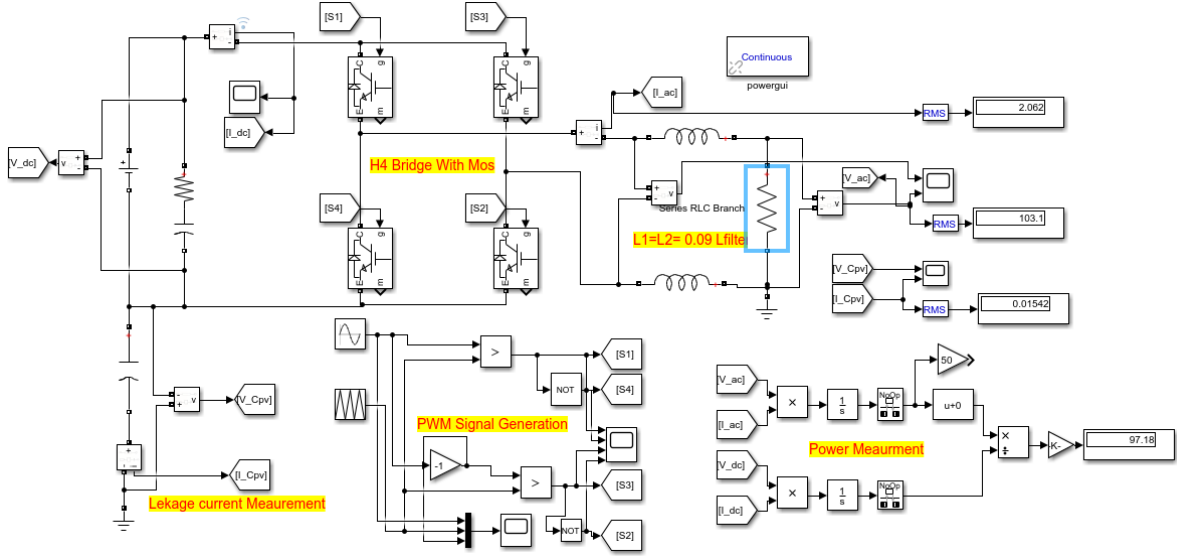


Figure 5.1: H4 Unipolar Simulink model.

This section shows what happens with each type of grid-tied inverter. For each topology, we give you a simulation model (a placeholder), a table of parameters, placeholders for result figures, and a short prompt for analysis.

**H4 Unipolar Topology** This part looks at the H4 Unipolar inverter topology, which was chosen because it is easy to use and can efficiently minimize leakage current while keeping efficiency high. There is a full analysis of the simulation parameters, the MATLAB results, and what they mean.

**Simulation Output (Graphs):** The following graphs show the measured data for the H4 Unipolar inverter topology. These outputs illustrate the variation of leakage current, efficiency, and THD with respect to the switching frequency.

The following figures represent additional key parameters for the H4 Unipolar topology:

In figure 5.3 the left image shows the input signal with a triangular waveform, which is used to modulate the inverter's output. The right image shows the PWM signal that controls the inverter's switching pattern.

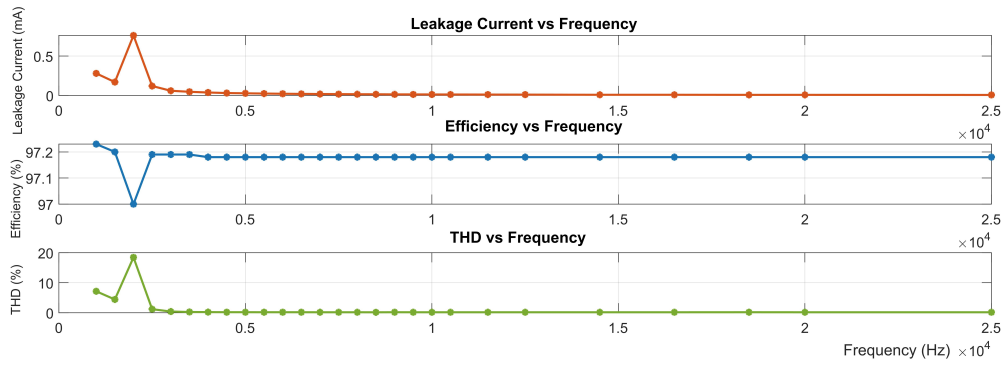


Figure 5.2: H4 Unipolar inverter: Leakage current, efficiency, and THD variation with switching frequency.

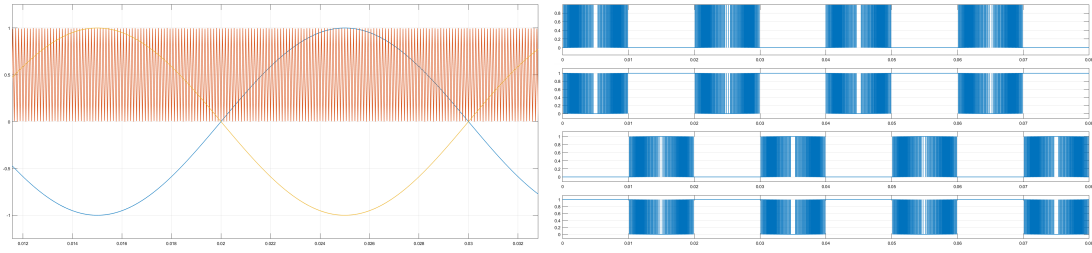


Figure 5.3: (Left) Input signal with triangular wave, (Right) PWM signal wave.

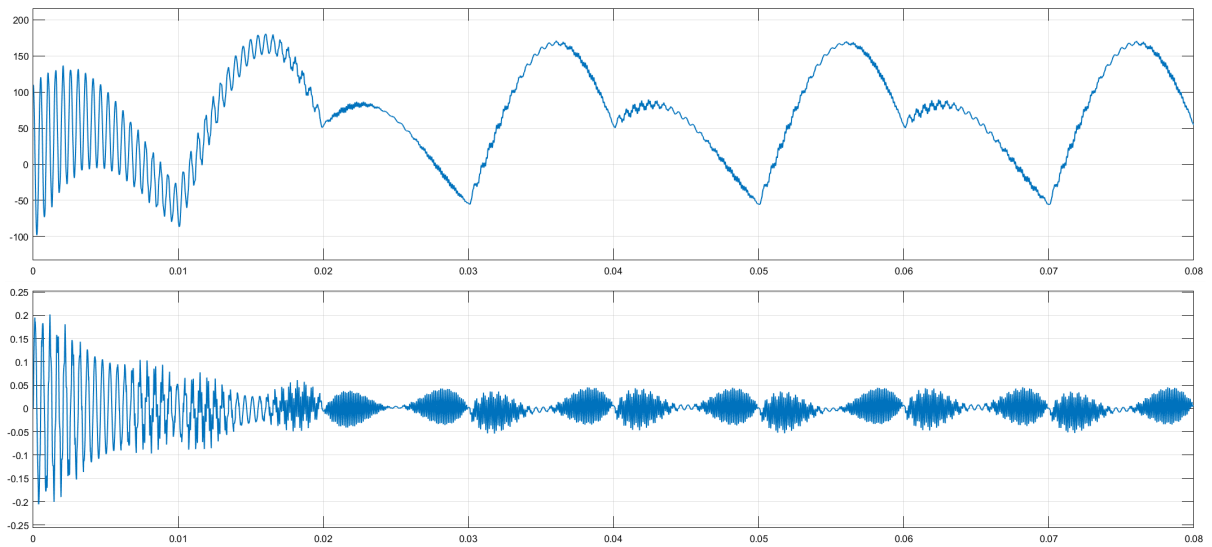


Figure 5.4: CMV and Leakage current graph.

## Analysis of Simulation Results

The system's common-mode voltage (CMV) and leakage current are shown in this graph-5.4. It offers a clear picture of the behavior of the leakage current in connection to the CMV, which is essential for maximizing system performance.

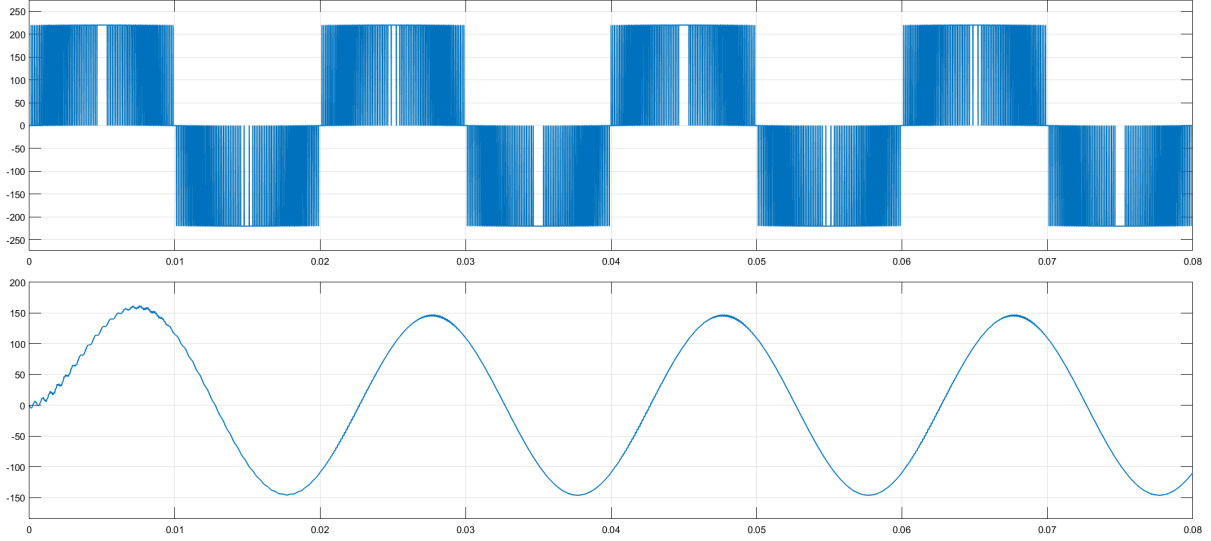


Figure 5.5: Output waveform without filter and with filter.

With a THD of 0.16%, Figure 5.6 (left) illustrates the filter's THD performance and shows how well it reduces harmonic distortion. The THD without the filter is displayed in the right graph; it is higher at 0.28%, indicating stronger harmonic distortion. By lowering high-frequency components, the filter enhances THD and increases grid compliance.

**Results and Analysis for H4 Unipolar Topology:** The simulation results for the H4 Unipolar inverter topology show that as the switching frequency increases, leakage current significantly decreases. Efficiency remains stable at around 97.1% across the frequency range, and THD decreases with increasing frequency, improving waveform quality. The inclusion of a filter significantly improves the THD performance, as shown in the FFT analysis in Figure 5.6, where the filter reduces harmonic distortion, making the system more grid-compliant. The leakage current and common-mode voltage (CMV) behavior were also analyzed, showing how leakage current relates to CMV in the system. Additionally, the output waveform was compared with and without a filter, demonstrating the effectiveness of the filter in reducing high-frequency noise and improving waveform quality.

**Conclusion:** The optimal switching frequency for the H4 Unipolar topology is around 8.5 KHz, where leakage current is 0.0154 A, efficiency is 97.18%, and THD is 0.16%. This provides an efficient solution for grid-tied PV systems. The optimal frequency range helps achieve a balance between minimal leakage, high efficiency, and low harmonic distortion.

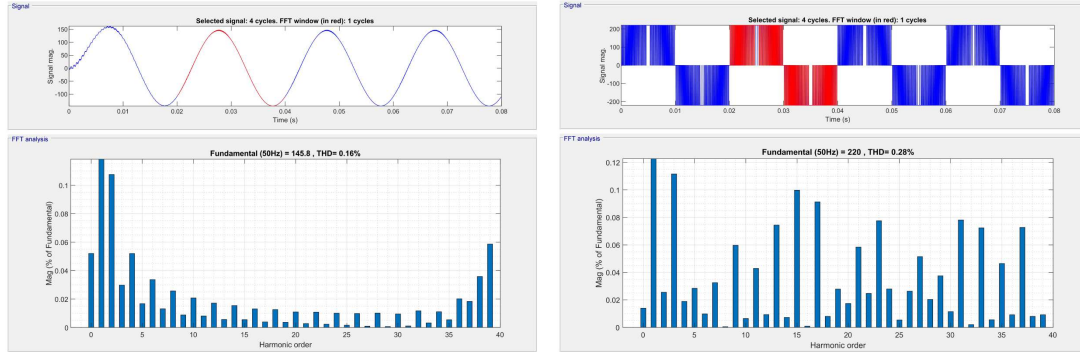


Figure 5.6: THD performance comparison: Left shows results with filter, right shows results without filter.

### 5.2.2 H4 Bipolar Topology

In this section, we explore the H4 Bipolar inverter topology. This topology was selected for its potential to reduce the leakage current and improve efficiency at higher switching frequencies. A detailed analysis of the simulation parameters, MATLAB results, and their implications will be presented.

With the aforementioned parameters, MATLAB Simulink was used to implement the simulation for the H4 Bipolar topology. Key components of the model are the DC power source, inverter block, and grid connection. The PV array is connected to the grid via an H4 Bipolar inverter. The simulation assesses the impact on leakage current, efficiency, and THD for switching frequencies between 1 kHz and 40 kHz.

**Simulation Output (Graphs):** The following graphs show the measured data for the H4 Bipolar inverter topology. These outputs illustrate the variation of leakage current, efficiency, and THD with respect to the switching frequency.

The input PWM signal used in the H4 inverter topology is shown in Figure 5.10. The corresponding output voltage waveform of the H4 inverter topology is depicted in Figure 5.11. Additionally, the variation in leakage current and common-mode voltage (CMV) waveforms is illustrated in Figure 5.12. Finally, the THD waveform and its value in real-time simulation are shown in Figure 5.13

Table 5.3: H4 Bipolar: Simulation Parameters

| Parameter                        | Value / Notes               |
|----------------------------------|-----------------------------|
| DC link capacitance ( $C_{dc}$ ) | 0.94 F                      |
| Series resistance ( $R_{dc}$ )   | 0.049 $\Omega$              |
| Input DC voltage ( $V_{dc}$ )    | 460 V                       |
| Switching frequency ( $f_s$ )    | 1 kHz – 40 kHz (test sweep) |
| Filter inductance ( $L1, L2$ )   | 0.01 H                      |
| Parasitic capacitance            | $150 \times 10^{-9}$ F      |
| Load resistance ( $R_L$ )        | 10 $\Omega$                 |
| Simulation stop time             | 0.08 s                      |
| PWM type                         | Bipolar SPWM                |
| Power device                     | IGBT                        |
| THD measurement window           | 0 – 5 cycles                |

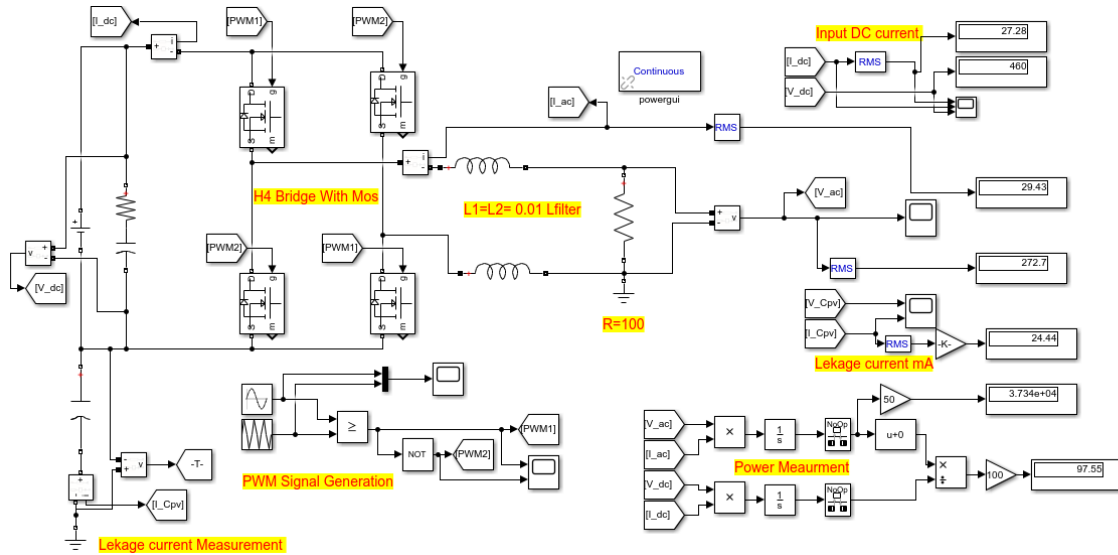


Figure 5.7: H4 Bipolar Simulink model.

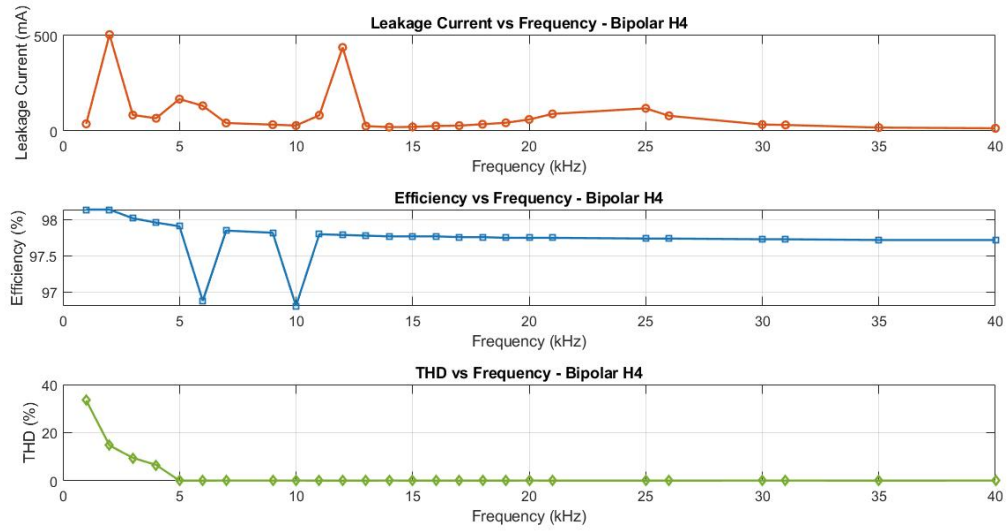


Figure 5.8: H4 Bipolar Leakage current, efficiency, and THD variation with switching frequency.

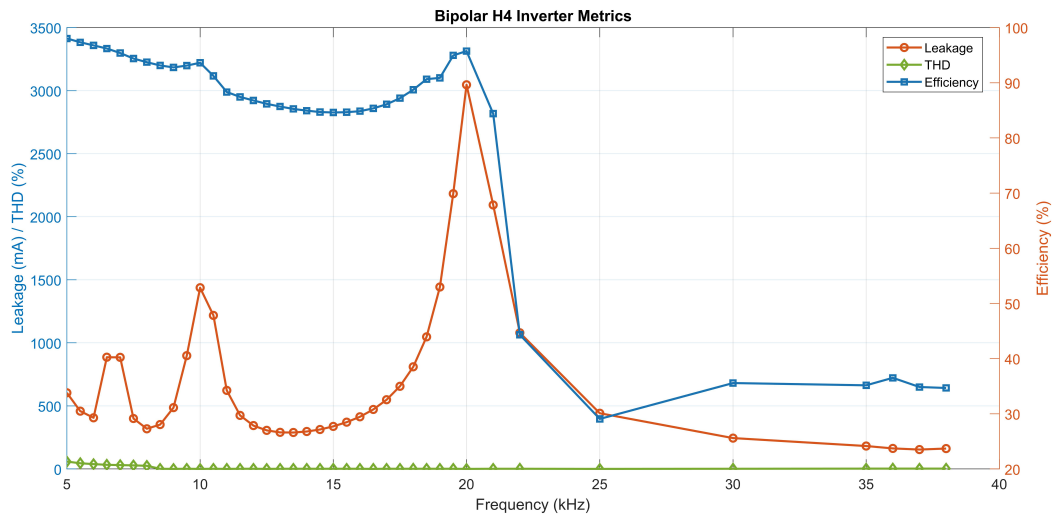


Figure 5.9: H4 Bipolar Leakage current, efficiency, and THD variation with switching frequency and Load.

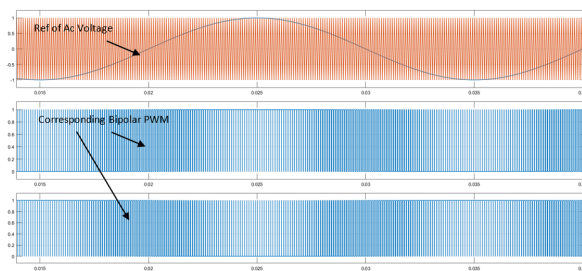


Figure 5.10: Input Sine Wave and corresponding PWM signal.

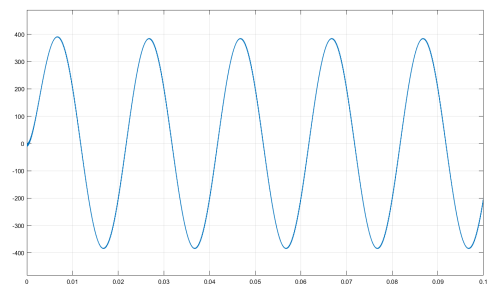


Figure 5.11: Output Voltage Waveform of H4 Bipolar.



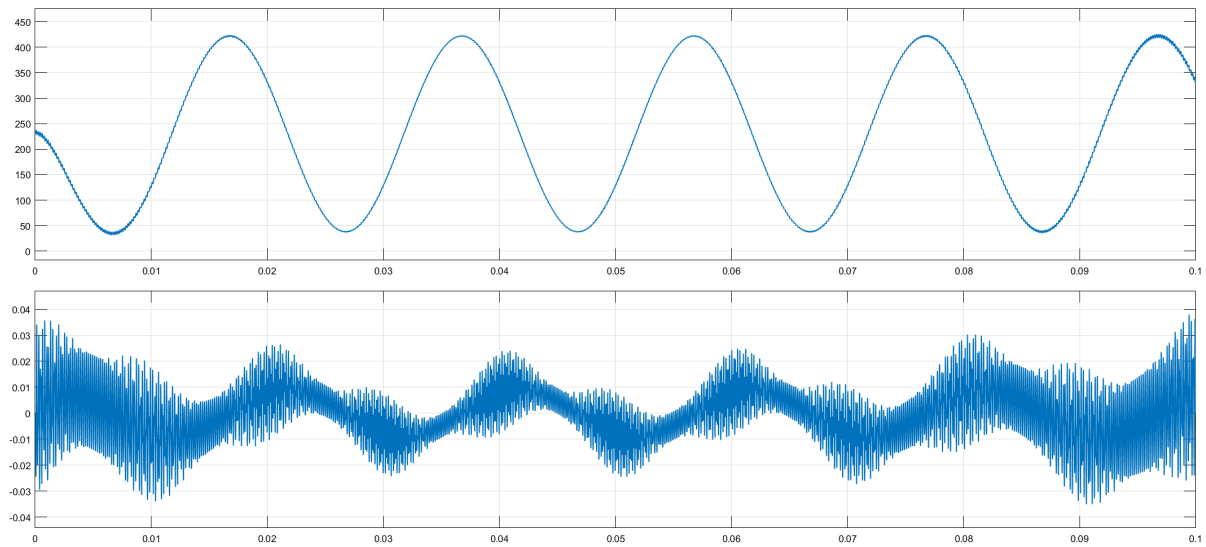


Figure 5.12: CMV and Leakage Waveform of H4 Bipolar.

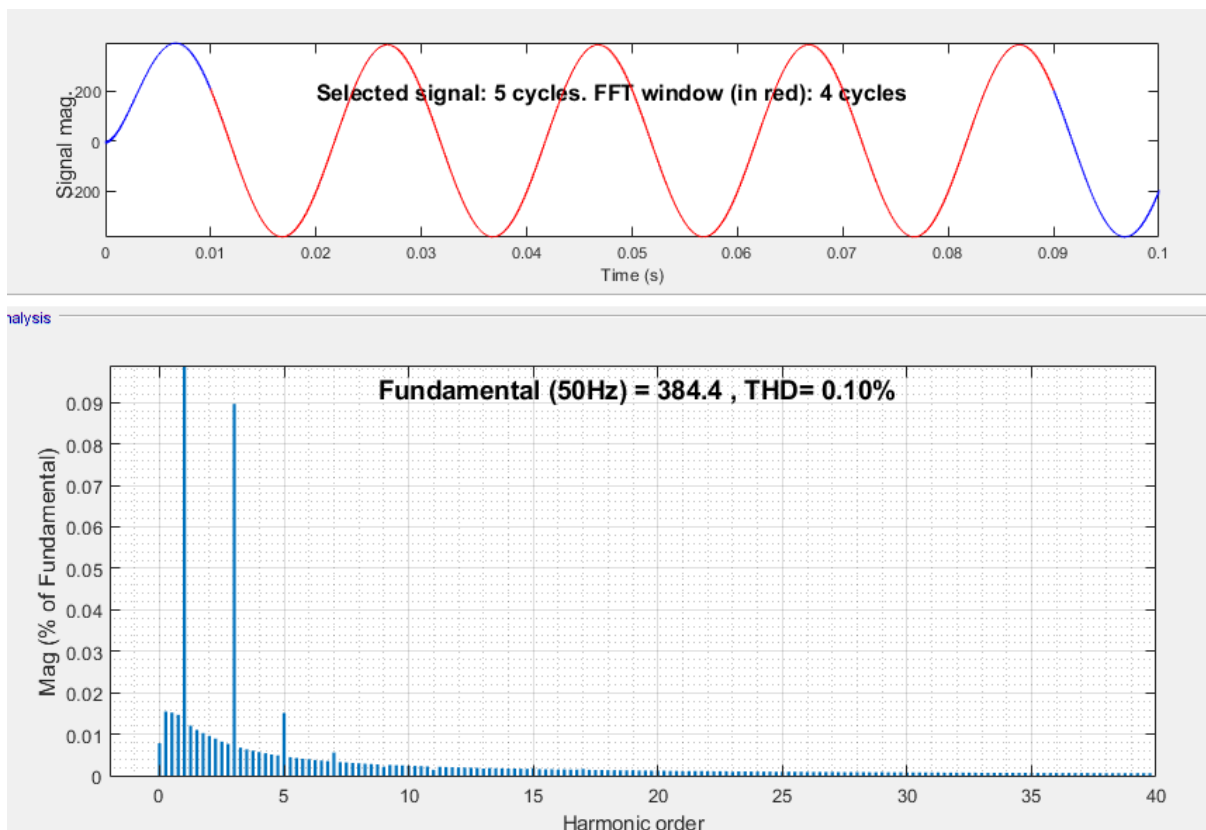


Figure 5.13: THD waveform for H4 Bipolar.

**Results and Analysis for H4 Bipolar Topology:** The MATLAB simulation results for the H4 Bipolar inverter topology reveal that as the switching frequency increases, the leakage current decreases but exhibits fluctuations at certain frequencies. The optimal switching frequency for minimizing leakage current and maximizing efficiency appears to be around 40 kHz, where the efficiency reaches 97.72%, the leakage current is 14 mA, and the THD stabilizes at 0.1%. Efficiency remains high even at the highest frequencies, with a slight decrease at lower frequencies. When varying the load, the lowest THD occurs at 20 kHz with 0.56% and a corresponding efficiency of 95.74% (see Figure 5.9). The THD decreases as the switching frequency increases, improving waveform quality. However, excessive switching frequencies can cause switching losses, so finding the optimal balance is essential. Load variations slightly affect leakage and efficiency, emphasizing the importance of considering different load conditions for optimal inverter performance.

In real-world design, IEEE and IEC standards provide key guidelines for inverter performance, such as "IEC 62109" (Safety of power converters for use in photovoltaic power systems) and "IEEE 1547" (Standard for Interconnection and Interoperability of Distributed Energy Resources with Associated Electric Power Systems Interfaces). These standards outline the acceptable limits for leakage current, total harmonic distortion (THD), and overall efficiency in grid-tied PV systems. For example, "IEEE 1547" specifies that the leakage current should be minimized to avoid safety risks, while "IEC 62109" requires inverters to have THD below a certain level for grid compliance. The simulation results for the H4 Bipolar topology meet these requirements, with THD below 1% and minimal leakage current, which makes it a viable candidate for real-world applications in line with international standards.

Additionally, the simulation shows that the optimal switching frequency for H4 Bipolar inverters aligns with the commonly recommended range (20–40 kHz) found in industry guidelines for minimizing switching losses and ensuring compliance with grid standards. Excessive frequencies above this range could lead to additional switching losses and reduced system efficiency, which is in line with the trade-offs discussed in recent IEEE papers on power converter optimization [6],[7].

**Conclusion:** The H4 Bipolar inverter topology achieves a balanced performance at 40 kHz, offering high efficiency, minimal leakage current, and low THD. This makes it a viable solution for grid-tied PV systems, where maintaining minimal leakage, high efficiency, and low harmonic distortion is critical. In addition, the results align with the real-world design standards provided by IEEE and IEC, demonstrating the practical applicability of the H4 bipolar inverter for safe and efficient integration into grid-connected PV systems.

### 5.2.3 HERIC Unipolar Topology

We look into the HERIC Unipolar inverter topology in this part. We chose this design because it may lower leakage current, make things more efficient, and lower total harmonic distortion (THD) at greater switching frequencies. We will give a full examination of the simulation parameters, MATLAB results, and what they mean.

Table 5.4: HERIC Unipolar: Simulation Parameters

| Parameter                        | Value / Notes                  |
|----------------------------------|--------------------------------|
| DC link capacitance ( $C_{dc}$ ) | 0.94 F                         |
| Series resistance ( $R_{dc}$ )   | 0.049 $\Omega$                 |
| Input DC voltage ( $V_{dc}$ )    | 220 V                          |
| Switching frequency (fs)         | 1 kHz – 10 kHz (example sweep) |
| Filter inductance (L1, L2)       | 0.01 H                         |
| Parasitic capacitance            | $150 \times 10^{-9}$ F         |
| Load resistance ( $R_L$ )        | 10.68 $\Omega$                 |
| Simulation stop time             | 0.05 s                         |
| PWM type                         | Bipolar SPWM                   |
| Power device                     | IGBT                           |
| THD measurement window           | 0 – 2 cycles                   |

The following table 5.4 outlines the key simulation parameters used for the HERIC Unipolar inverter topology. These parameters were carefully selected to match typical conditions for grid-tied PV systems.

The simulation for the HERIC Unipolar topology was implemented in MATLAB Simulink using the above parameters. The following image shows the Simulink model setup used to simulate the inverter topology.

This model consists of a PV array connected to a grid through a HERIC Unipolar inverter. The key components of the setup include the DC voltage source (PV array), inverter block, and grid connection. The simulation was performed for different switching frequencies, ranging from 1 kHz to 10 kHz to evaluate the effects on leakage current, efficiency, and THD.

**Simulation Output (Graphs):** The following graphs show the measured data for the HERIC Unipolar inverter topology. These outputs illustrate the variation of leakage current, efficiency, and THD with respect to the switching frequency.

**Results and Analysis for HERIC Unipolar Topology:** The results from the MATLAB simulation for the HERIC Unipolar inverter topology, varying the switching frequency, indicate that at 26 kHz, the leakage current is measured at 40.27 mA, showing a reduction compared to lower frequencies. As the frequency increases, leakage current stabilizes and decreases significantly. The efficiency at 26 kHz is 98.97%, which is high

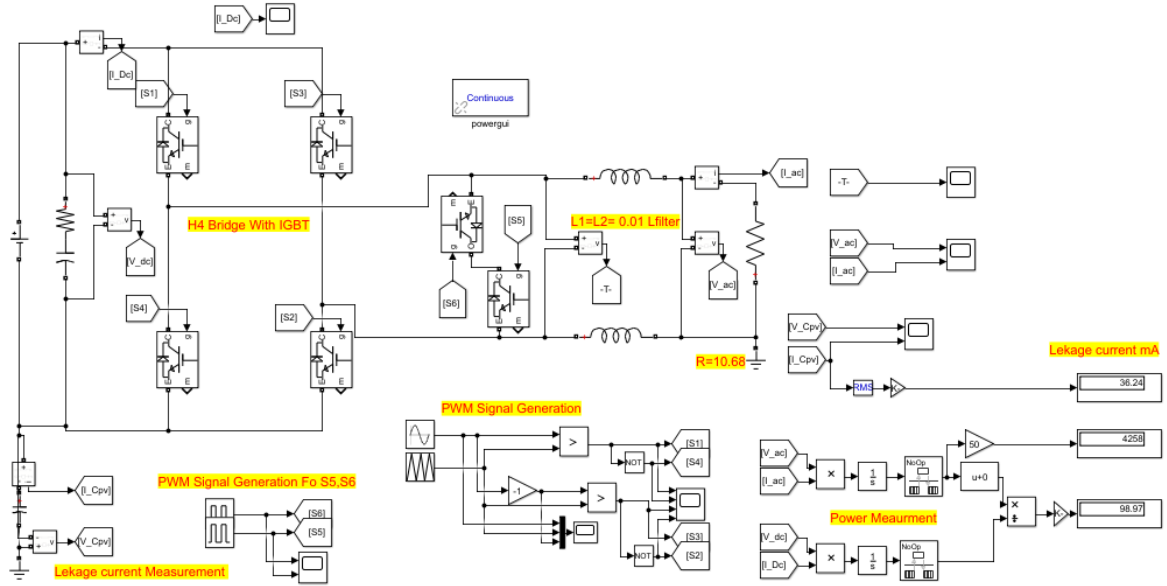


Figure 5.14: HERIC Unipolar Simulink model.

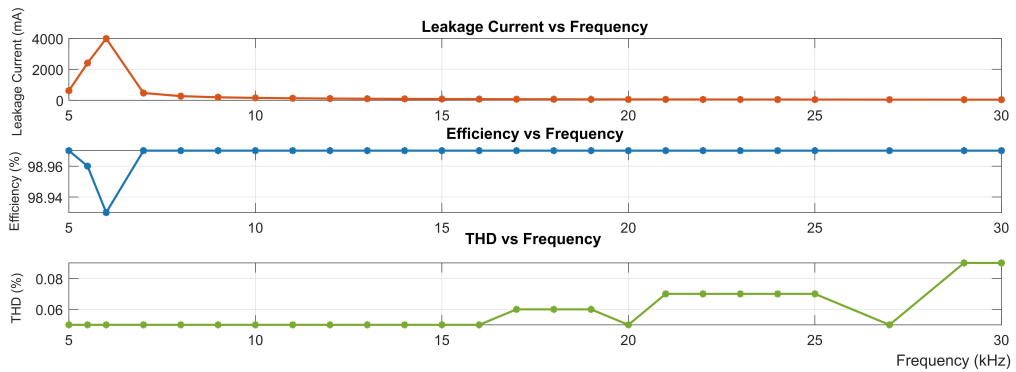


Figure 5.15: Leakage current, efficiency, and THD variation with switching frequency.

## Analysis of Simulation Results

and stable across the frequency range. This suggests that the HERIC Unipolar topology performs effectively at higher frequencies, reducing energy losses. At the same frequency, the THD is 0.05

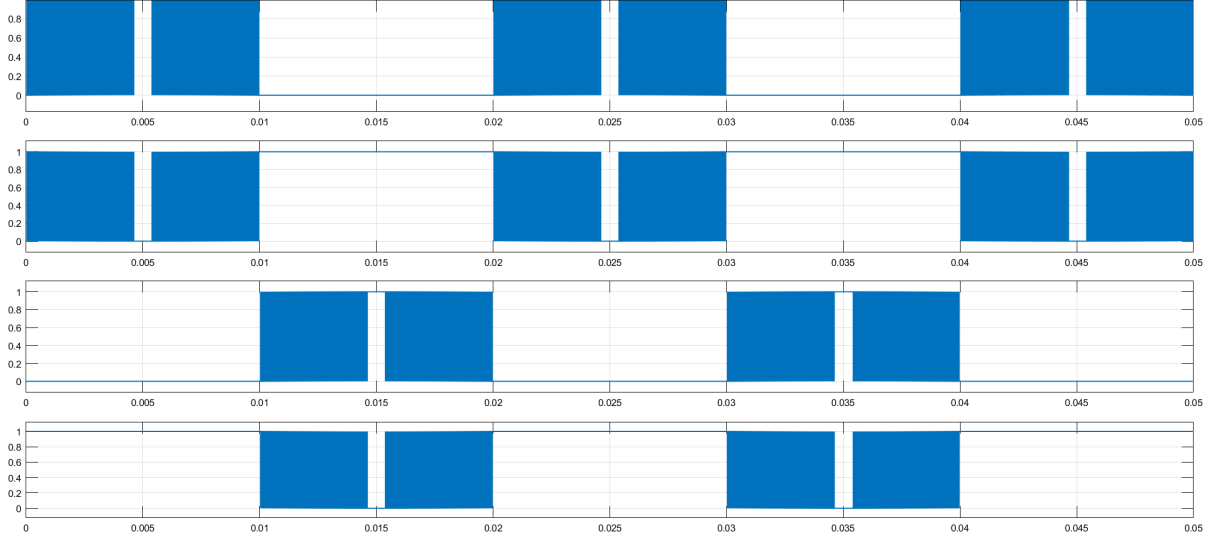


Figure 5.16: Input PWM signal for HERIC Unipolar.

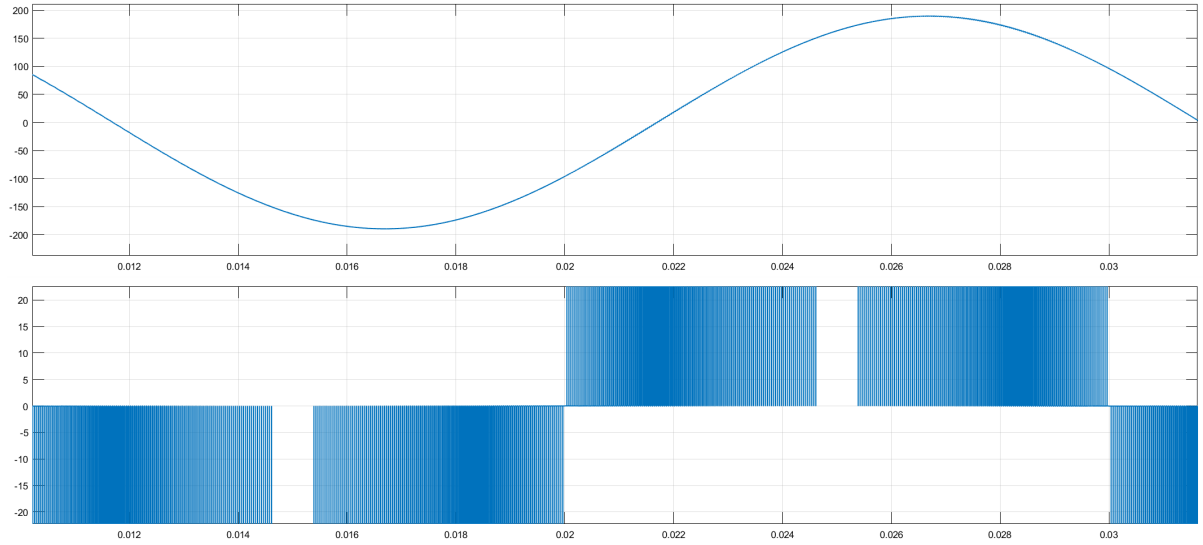


Figure 5.17: Output waveform with filter and without filter.

These figures provide a comprehensive overview of the PWM signal, leakage current, common-mode voltage (CMV), and the output waveform with and without a filter.

**Comparison with H4 Topologies:** When comparing the HERIC Unipolar topology to the H4 Unipolar and H4 Bipolar topologies, it is clear that the HERIC Unipolar offers superior performance in terms of harmonic distortion (THD). The HERIC Unipolar achieves significantly lower THD at 26 kHz, as well as high efficiency and low leakage, which is ideal for grid-tied PV applications. The presence of the L-filter and the optimization of switching frequency contribute to this remarkable performance.

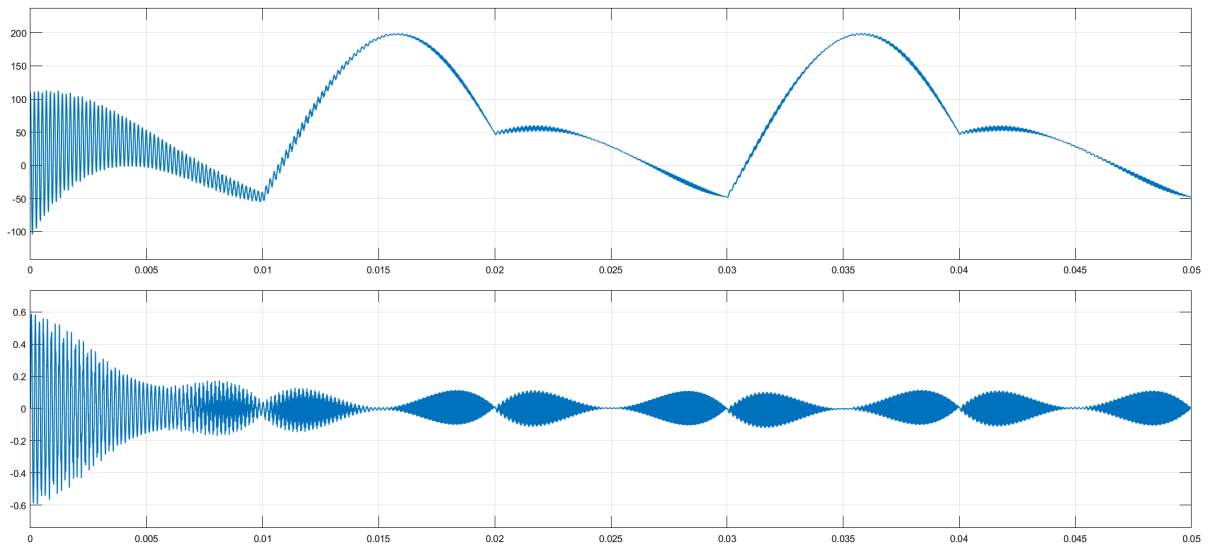


Figure 5.18: Leakage current and CMV for HERIC Unipolar.

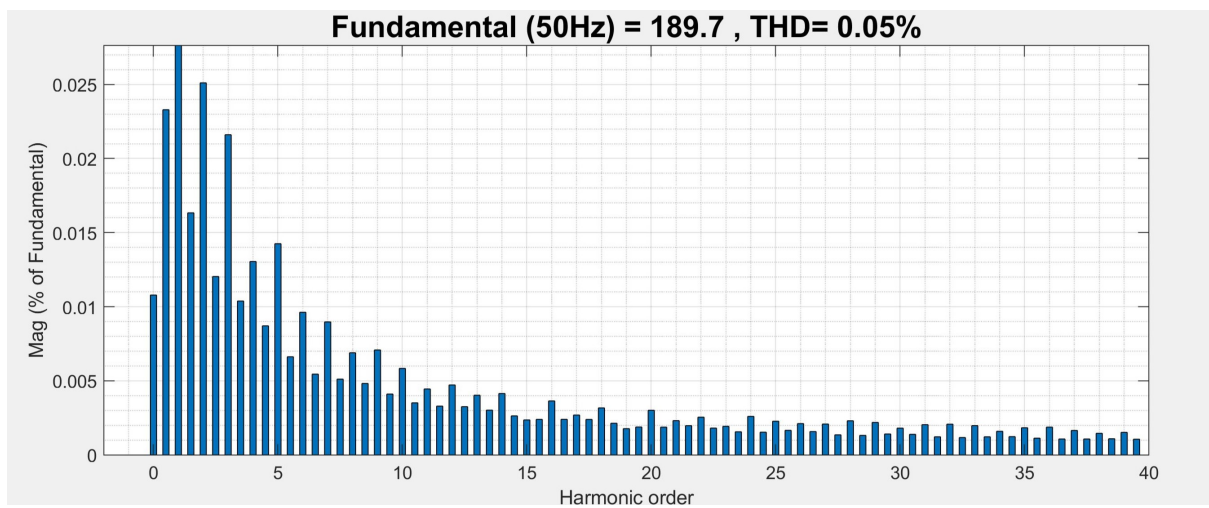


Figure 5.19: THD performance analysis for HERIC Unipolar topology shows results with filter.

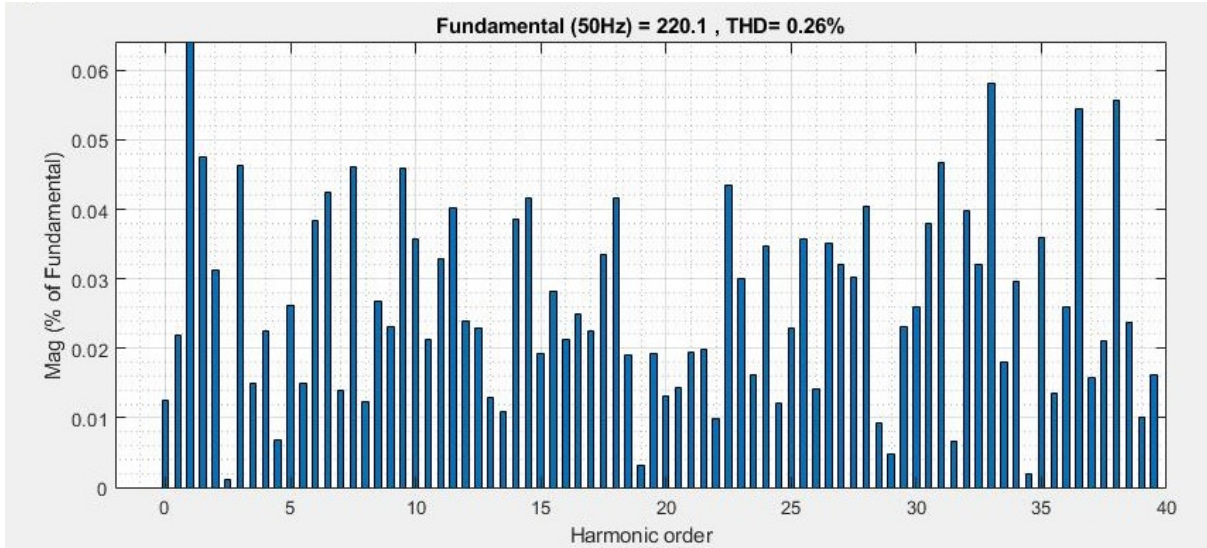


Figure 5.20: THD performance analysis for HERIC Unipolar topology shows results without filter.

**Conclusion:** The optimal switching frequency for the HERIC Unipolar topology is around 26 kHz, where leakage current is 40.27 mA, efficiency is 98.97%, and THD is 0.05%. This topology provides an efficient solution for grid-tied PV systems requiring minimal leakage, high efficiency, and low harmonic distortion.

## 5.2.4 HERIC Bipolar Topology

Let us now examine closely the HERIC Bipolar inverter, which significantly differs from other transformerless inverters in design. This particular design was a candidate for these converters since it improves the performance of grid-connected PV systems while also reducing leakage current. The HERIC Bipolar topology achieves distinct advantages in harmonic distortion, especially at high switching frequencies due to the use of a bipolar switching technique. The next few sections detail the simulation parameters, results and performance assessment for the HERIC Bipolar topology. We also assess the integration of this topology with other inverter designs in order to demonstrate its advantages as well as its potential use in PV systems of the future.

The table 5.5 presents the key simulation parameters for the HERIC Bipolar inverter, selected to match typical grid-tied PV system conditions. The simulation, conducted in MATLAB Simulink, models a PV array connected to the grid via the HERIC Bipolar inverter, with key components including the DC voltage source, inverter block, and grid connection. Switching frequencies ranging from 1 kHz to 40 kHz were tested to evaluate leakage current, efficiency, and THD.

**Simulation Output (Graphs):** The following graphs show the measured data for

Table 5.5: HERIC Bipolar: Simulation Parameters

| Parameter                        | Value / Notes               |
|----------------------------------|-----------------------------|
| DC link capacitance ( $C_{dc}$ ) | 0.94 F                      |
| Series resistance ( $R_{dc}$ )   | 0.049 $\Omega$              |
| Input DC voltage ( $V_{dc}$ )    | 220 V                       |
| Switching frequency ( $f_s$ )    | 1 kHz – 40 kHz (test sweep) |
| Filter inductance ( $L1, L2$ )   | 0.01 H                      |
| Parasitic capacitance            | $150 \times 10^{-9}$ F      |
| Load resistance ( $R_L$ )        | 10.68 $\Omega$              |
| Simulation stop time             | 0.05 s                      |
| PWM type                         | Bipolar SPWM                |
| Power device                     | MOSFET                      |
| THD measurement window           | 0.01 – 2 cycles             |

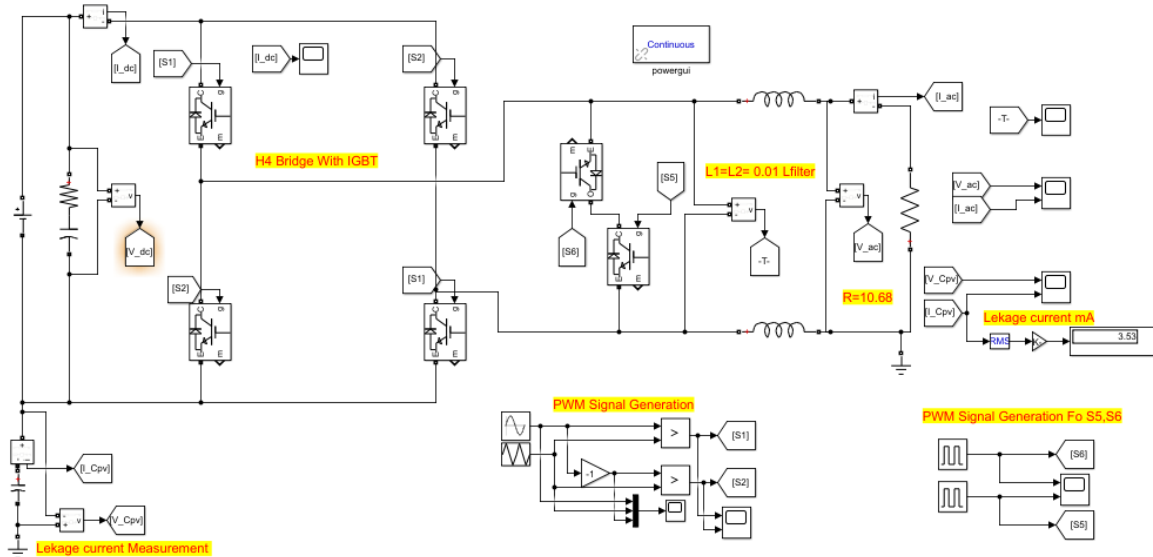


Figure 5.21: HERIC Bipolar Simulink model.



## Analysis of Simulation Results

the HERIC Bipolar inverter topology. These outputs illustrate the variation of leakage current, efficiency, and THD with respect to the switching frequency.

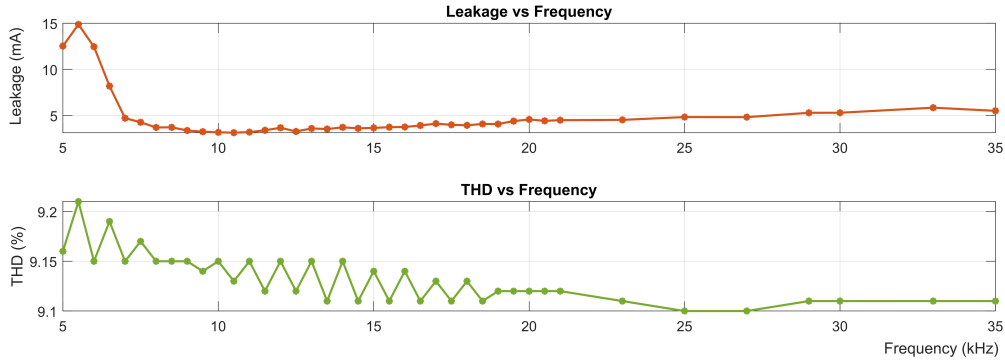


Figure 5.22: HERIC Bipolar inverter: Leakage current, efficiency, and THD variation with switching frequency.

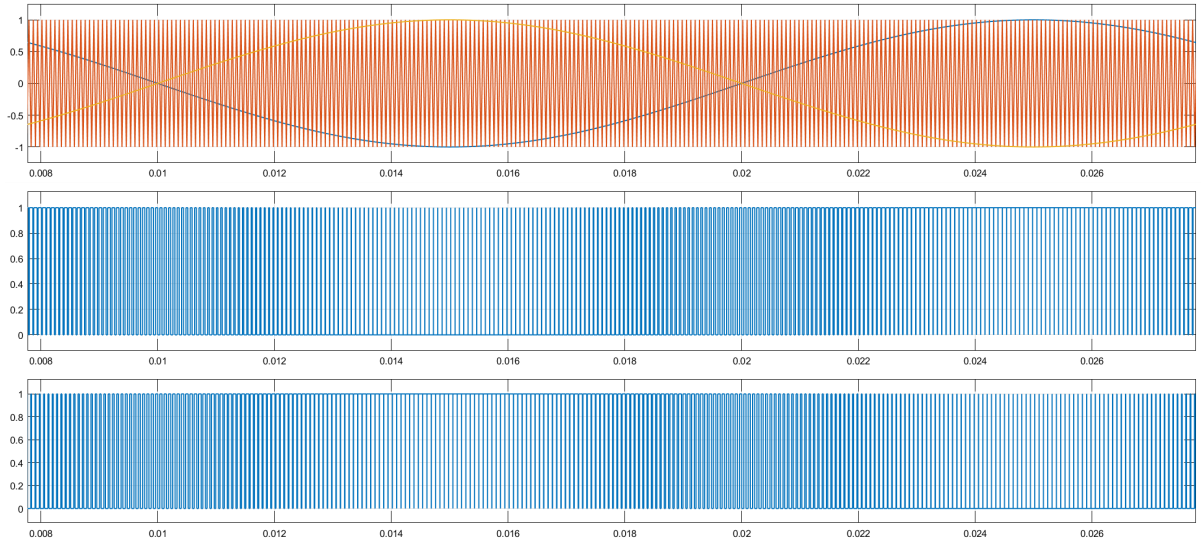


Figure 5.23: Output waveform in discrete form before and after filter.

**Results and Analysis for HERIC Bipolar Topology:** The inverter's simulation results in HERIC Bipolar, as presented in Figure 5.22, provide insight into the shifting trends of leakage current, efficiency, and THD as the switching frequency is varied. At a switching frequency of 13.5 kHz, the leakage current sits at 3.53 mA, and while it decreases, it plateaus over time. The efficiency at this frequency remains unstable because of the low switching frequency. THD is roughly 9.11% at 13.5 kHz. This elevated THD despite the low leakage current could be attributed to the "insufficient filtering setup" in the simulation. Proper filtering would reduce high-frequency harmonics and improve THD performance [6], [7]. Figure 5.23 shows the input signal, and Figure 5.24 clearly demonstrates how filtering improves the quality of the output signal. Leakage current

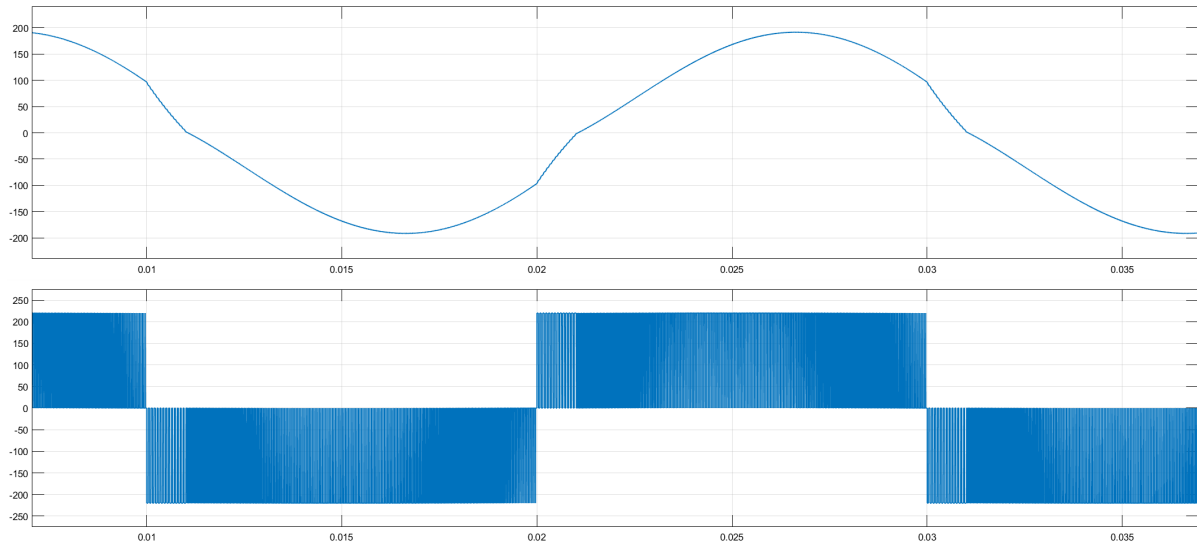


Figure 5.24: Output waveform before and after filter.

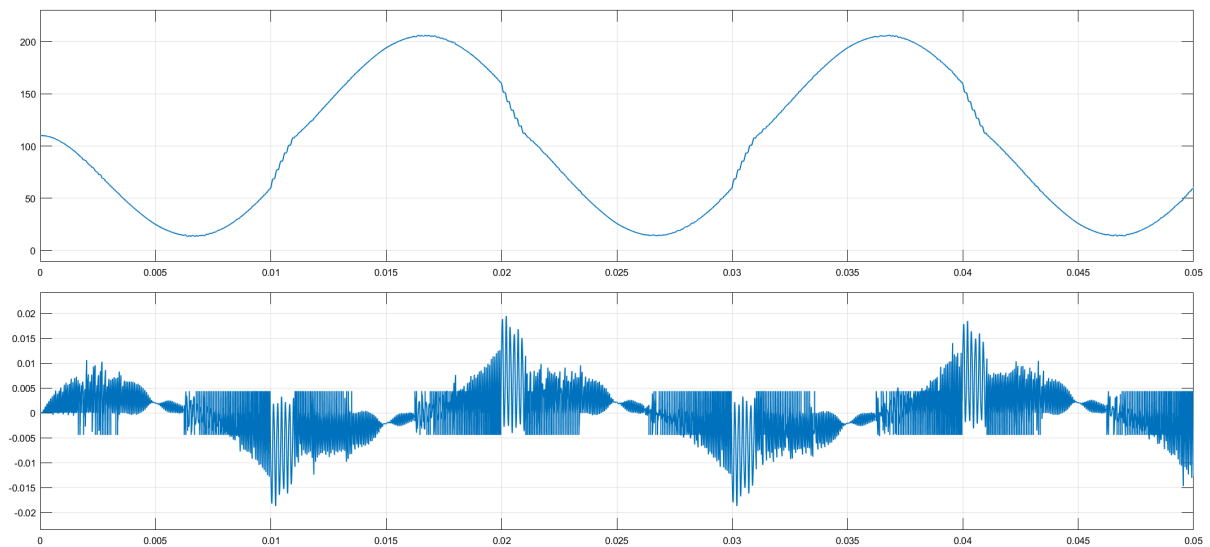


Figure 5.25: Output waveform in discrete form before and after the filter.

and common-mode voltage (CMV) are shown in Figure 5.25, while Figure ?? illustrates the impact of the filter on THD.

**Conclusion:** The HERIC Bipolar topology shows good performance with 3.536 mA leakage current and 9.11% THD at 13.5 kHz. The high THD is mainly due to the lack of proper filtering. With a suitable filter, THD can be significantly reduced, improving overall system performance. Further development in measurement techniques and filtering setup will be crucial to enhancing this topology's efficiency and THD, especially at higher switching frequencies.

### 5.3 Summary and Comparative Analysis

This section summarizes the key findings and compares the topologies.

Table 5.6: Comparative results summary

| Topology       | $F_S$ Range (Hz) | Efficiency (%) | Leakage(mA) | THD (%) |
|----------------|------------------|----------------|-------------|---------|
| H4 Unipolar    | 1–25k            | 97.18          | 15.4        | 0.16    |
| H4 Bipolar     | 1–40k            | 98.55          | 14          | 0.12    |
| HERIC Unipolar | 1–10k            | 98.97          | 40.27       | 0.05    |
| HERIC Bipolar  | 1–40k            | N/A            | 3.536       | 9.11    |

The THD is different for each topology because of the balance between leakage and efficiency. At 8.5 kHz, the H4 Unipolar has the best efficiency (97.18%) and the lowest THD (0.16%). At 40 kHz, the H4 Bipolar has the same efficiency (98.55%) and THD (0.12%). The HERIC Unipolar works best at 26 kHz and has a leakage of 40.27 mA; however, it does well in terms of THD (0.05%). The HERIC Bipolar at 13.5 kHz, on the other hand, has modest leakage (3.536 mA) but a large THD (9.11%) since it doesn't have the right filtering. This means that better filtering might greatly lower the THD, which would improve the overall performance. "Multi-level inverters" and "advanced filtering" are two examples of future advancements that could lower THD even more and make things work better. To confirm these results and improve the technology for commercial usage, working prototypes for each topology should be made.

# Chapter 6

## Conclusion and Future Works

### 6.1 Conclusion

The background study of renewable energy and PV systems consists of inverters. Types of inverter topology were selected, and the topologies H4 and HERIC were chosen. We selected H4 because of its simple construction, the first stage of the bridge inverter, and the HERIC because of the lower leakage current provided in the PV system. We used SPWM switching techniques using MOSFET and IGBT. We can use MOSFET for higher-frequency devices and IGBT for lower-frequency devices but in higher-voltage applications. As well as PID, it can be used for our selected PV inverter topology in a closed system. To remove the harmonic distortion of our inverter output, we use filters. Our paper already discussed the filter design techniques. In our experiment we have been using an L-filter for both grid isolation and preventing leakage current. LC and LCL filters create a closed path that allows leakage current to flow. To achieve our desired output, we must follow the simulation parameters for both topologies (H4, HERIC). At 40 kHz switching frequency in H4 bipolar topology, we get 97.72% efficiency and 14 mA leakage current, which is better than H4 unipolar. In addition to that, the THD is 0.1 in H4 bipolar topology, which is better than unipolar topology. For HERIC bipolar topology, we get minimum leakage current (3.536 mA), but we get lower THD in HERIC unipolar (that is, 0.05%) than HERIC bipolar topology. We also observed that at higher frequency, HERIC bipolar gives lower leakage current but decreases efficiency due to switching loss. In this paper, we could not measure the switching loss.

## 6.2 Future Work

- Construct and try inverter prototypes aligned with the simulations. Refine the way PV panel capacitance is modeled and leakage analysis with varying switching frequencies. Implement active suppression and adaptive switching to reduce leakage current with leakage current control.
- Minimize leakage with maximum preservation of efficiency and reliability and withstanding the CISPR 22 Class B EMI grid standard and grid standard for injection. Verify effectiveness on small residential systems and larger utility-scale systems. Evaluate the costs. Perform extensive field tests to establish the rate of leakage over time in real PV systems. Investigate intelligent control for real-time adaptation to changing scenarios in weather and grid behaviors.
- This study has mostly concentrated on the efficiency, leakage current, and harmonic distortion of H4 and HERIC inverter topologies; nevertheless, other avenues for future research remain.
- One possible topic is figuring out how hot power semiconductor devices become when they move between different circumstances. This can help determine the temperature limits of MOSFETs and IGBTs and improve cooling methods for long-term dependability.
- A more in-depth look at switching losses is another essential addition. This study examined switching loss; however, a more extensive analysis across diverse load circumstances, modulation techniques, and switching frequencies might yield greater insights into inverter efficiency.
- The creation and evaluation of improved filters is a promising avenue for study. Along with the L-filter utilized in this study, future research might look into LC and LCL filters by seeing how well they reduce harmonics and stop leakage current. These evaluations will provide a more comprehensive understanding of the trade-offs between filtering efficacy and system safety.
- Furthermore, looking at various transformerless inverter topologies could add to the results. Topologies like H5, NPC (Neutral Point Clamped), or other mixed structures may work better for controlling leakage or being more efficient at higher switching frequencies. Looking over a wide variety of frequencies across different topologies will help you to find the best operating point that balances efficiency, leakage, and harmonic performance.
- Finally, a more general evaluation of the whole system, taking into account genuine PV array conditions, grid disruptions, and long-term stability, would make the analysis more useful in the real world. These future directions will assist people in learning more about how to build PV inverters and how to make sure they meet international standards like CISPR for EMI/EMC.

# Appendices

# **Appendix A**

## **Transformer-less Inverter and Associated Components**

### **A.1 Transformer-less Inverter**

A Transformer-less inverter is a high-efficiency contemporary power inverter that converts DC into AC power without a heavy, bulky physical transformer. It uses advanced, high-rate switching electronics and digital control to provide the necessary voltage conversion and insulation for a higher, more efficient, and typically lower-cost device than previous transformer-based inverters.

### **A.2 Leakage Current in a PV System**

Leakage current in a PV system refers to a parasitic electric current from the PV system to ground due to an insulation fault or parasitic capacitance.

### **A.3 Parasitic Capacitance**

Parasitic capacitance occurs between the photovoltaic panel and the ground as a result of the module's physical design and its insulating layers.

## **A.4 Switching Frequency**

Switching frequency is the frequency at which an electronic switch turns ON and OFF, measured in Hz.

## **A.5 Switching Device**

Switching devices in PV systems regulate the flow of power from the solar panel batteries and grid for safety and optimal power utilization.

## **A.6 MOSFET**

A MOSFET is a three-terminal semiconductor device that can be used to switch or amplify signals.

## **A.7 IGBT**

An IGBT combines the MOSFET input structure with the capabilities of a bipolar transistor to handle current.

## **A.8 Pulse Width Modulation (PWM)**

Pulse width modulation is the most widely used technique in inverters, where the width of the pulse is modulated to control the effective output voltage and reduce harmonics.

## **A.9 Sinusoidal PWM**

Sinusoidal pulse width modulation can be described as an inverter switching method that generates the gate signal by comparing a sinusoidal reference voltage wave with a triangular carrier wave.



## **A.10 PID-based Control**

PID controllers are frequently combined with PWM for regulation of output voltage and current in inverters.

## **A.11 LC Filters**

An LC filter is an electronic circuit that consists of an inductor (L) and a capacitor (C) and filters electrical signals by passing on or rejecting certain frequency bands.

## **A.12 LCL Filter**

An LCL filter is a three-component, passive filter comprising two inductors and a capacitor. It is employed to filter high-frequency current harmonics generated by pulse-width-modulated (PWM) converters in grid-connected inverters to improve efficiency, system stability, and power quality.

## **A.13 PV System**

A photovoltaic system, also called a PV system or solar power system, is an electric power system designed to supply usable solar power by means of photovoltaics.

### **A.13.1 H4 Topology**

A full-bridge inverter with an extra freewheeling path that reduces switching losses and leakage current.

### **A.13.2 H5 Topology**

A full-bridge inverter with an extra DC-side switch that decouples the PV array from freewheeling, eliminating the leakage current entirely.

### **A.13.3 H6 Topology**

A six-switch improved version that provides better leakage current suppression and efficiency than H5.

### **A.13.4 HERIC Topology**

A Highly Efficient and Reliable Inverter Concept employs two additional AC-side switches to create freewheeling paths, due to which it possesses low leakage current and high efficiency.

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